

Eigenvalues of Tetrahedral sums of Hermitian matrices

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Abstract

TC: This is the MAIN draft for the TETRASPECTRA project. Older drafts are located in the folder titled “old” and will NOT compile properly.

Horn’s problem is concerned with characterizing the eigenvalues (α, β, γ) of a triple of Hermitian matrices (A, B, C) satisfying the constraint $A+B = C$. The development of a complete solution to Horn’s problem, due in large part to breakthroughs by A. Klyachko, A. Knutson, and T. Tao, has led to deep connections between eigenvalues of Hermitian matrices, tensor product invariants of matrix groups & Littlewood-Richardson coefficients, geometric invariant theory and the intersection theory of Schubert varieties via cohomology.

Here, we define and investigate a generalization of Horn’s problem which we refer to as the *tetrahedral Horn’s problem*. The tetrahedral Horn’s problem is concerned with characterizing the eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ of a sextuple of Hermitian matrices (A, B, C, D, E, F) satisfying the constraints $A+B = C$, $B+D = F$, $D+C = E$ and $A+F = E$. In this paper, we describe the scaling, shifting and oriented tetrahedral symmetries of the tetrahedral Horn’s problem, as well as conjecture the existence of additional symmetries generalizing the Regge symmetries present in the case of 2×2 Hermitian matrices. Furthermore, unlike the solution to Horn’s problem, we show that the space of admissible sextuples of eigenvalues is neither convex nor a polytope. We demonstrate this qualitative difference explicitly in the case of 2×2 Hermitian matrices.

For the general case of $n \times n$ Hermitian matrices, we collect a handful of previously known matrix inequalities for sums of positive semidefinite matrices and reinterpret them as necessary conditions for the tetrahedral Horn’s problem. For a complete characterization, we provide a direct proof, using eigenvalue estimation techniques from quantum information theory, that the admissibility of a given sextuple of eigenvalues is encoded in the semiclassical asymptotics of the generalized Wigner $6j$ symbols for the unitary group $U(n)$. Our result generalizes the well-known correspondence between admissible eigenvalues for Horn’s problem and Littlewood-Richardson coefficients, and partially generalizes the well-known semiclassical analysis of J. Roberts for the case of $n = 2$.

Finally, we prove an upper bound on general $6j$ symbols using the Cauchy-Schwarz inequality which, in the $n \rightarrow \infty$ asymptotic limit, gives rise to a novel proof of a known strongly subadditive inequality involving unnormalized entropies.

Contents

1	Introduction	2
2	Tetrahedral eigenvalues & the tetrahedral Horn problem	5
3	Symmetries of tetrahedral eigenvalues	9
4	Eigenvalues of blocks of positive semidefinite matrices	13
5	Inequalities for tetrahedral eigenvalues	16
5.1	Trigonal Horn inequalities	17
5.2	Submodular inequalities	17
5.3	The CŠW inequality	21
6	Representation theory & eigenvalue estimation	22
6.1	Schur-Weyl duality & eigenvalue estimation	22
6.2	Littlewood-Richardson coefficients & trigonal eigenvalues	23
6.3	Generalized $\mathbf{6j}$ symbols & tetrahedral eigenvalues	23
6.4	The large n limit & an entropic inequality	30
7	OLD CONTENT	33
7.1	Littlewood-Richardson coefficients & $\mathbf{6j}$ symbols	33
7.2	Littlewood-Richardson coefficients & the trigonal Horn problem	35
A	Multinomial distribution inequalities	39

1 Introduction

There are a number of operations involving the transformation of Hermitian matrices under which it is possible to track the effect had of the corresponding sets of eigenvalues. For example, the eigenvalues, $\text{eig}(X)$, of a Hermitian matrix, X , are invariant under conjugation by unitary matrices, $\text{eig}(X) = \text{eig}(UXU^\dagger)$, covariant under taking positive powers $\text{eig}(X^p) = \text{eig}(X)^p$, covariant under multiplication by real numbers, $\text{eig}(rX) = r \text{eig}(X)$, and covariant under the addition of scalar matrices, $\text{eig}(X + rI) = \text{eig}(X) + (r, \dots, r)$. The operation of adding together Hermitian matrices is not an operation of this type. Specifically, the eigenvalues $\gamma = \text{eig}(A + B)$ of the sum of two matrices cannot be determined from the eigenvalues $\alpha = \text{eig}(A)$ and $\beta = \text{eig}(B)$ of the summands *alone*. Stated differently, to determine the eigenvalues involved, additional knowledge about the eigenvectors involved is typically required. Nevertheless, there are numerous mathematical situations wherein knowledge about the eigenvectors involved is lacking, yet it remains desirable to determine the set of admissible eigenvalues of a sum given the eigenvalues of the summands.

The problem of determining the set (α, β, γ) of admissible eigenvalues has been investigated for over a century yet has come to be known as *Horn's problem* after Horn who, in 1962, conjectured a solution to the problem in the form of an inductively defined, finite set of homogeneous linear inequalities. These inequalities, now known as the *Horn inequalities* describe a polytope of admissible values for (α, β, γ) [Hor62]. In 1998, Klyachko provided a solution to Horn's problem [Kly98] which, when combined with a proof of the so-called saturation conjecture, due to Knutson and Tao [KT99], constitutes a proof of Horn's original conjecture [Ful98].

For applications of Horn's problem to quantum information theory and Nielsen's theorem [Nie99], see the article due to Daftuar and Hayden [DH05].

In parallel to Horn's problem, numerous variations and extensions of Horn's problem have been explored by various authors. For example, numerous authors have considered the admissible eigenvalues of unitary matrices and their product [AW97; Bel08; BK16], or the admissible singular values of matrices and their product [Spe05; TT71]. In 2000, it was shown by [Kly00] that *logarithms* of the admissible singular values of non-singular matrices and their product are identical to the admissible eigenvalues of Hermitian matrices and their sum, thus proving a conjecture due to Thompson [Kly00]. This correspondence between the singular

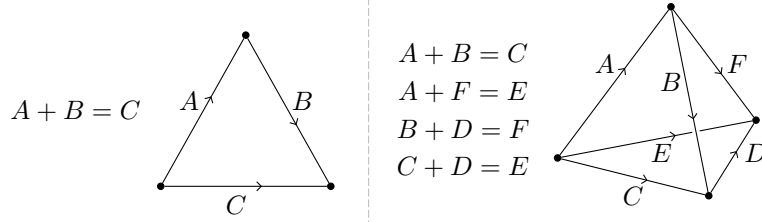


Figure 1: (Left) The *triangular* Horn's problem subjects three Hermitian matrices, (A, B, C) , to the constraint $A + B = C$ forming a triangle in the vector space of Hermitian matrices. (Right) The *tetrahedral* Horn's problem subjects six Hermitian matrices, (A, B, C, D, E, F) , to a quadruple of constraints forming a tetrahedron in the vector space of Hermitian matrices. fig:tetrahedron_labelling_scheme

values of a product and the eigenvalues of sum has been generalized substantially by Alekseev, Meinrenken, and Woodward [AMW01]. For a pedagogical and historical introduction to Horn's problem, we recommend the review articles of Bhatia [Bha01], Knutson [Knu00], and Fulton [Ful00].

Recent developments surrounding Horn's problem not covered by the aforementioned reviews include the development of the geometric method by Sherman [She15] and Berline, Vergne, and Walter [BVW18] originally pioneered by the geometric proofs due to Belkale [Bel02; Bel05], as well the characterization and computation of the induced distribution over the eigenvalues of the sum given random matrices with fixed spectra in a series of papers by Coquereaux and Zuber [CZ18; CZ18; CMZ19; Zub18].

In this paper, we consider a generalization of Horn's problem previously considered only briefly in a paper due to Christandl, Şahinoğlu, and Walter [CSW18]. While Horn's problem aims to characterize the eigenvalues (α, β, γ) of a triple of $n \times n$ Hermitian matrices (A, B, C) which are subjected to the constraint $A + B = C$, the generalization consider here aims to characterize the eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ of a *sextuple* of $n \times n$ matrices (A, B, C, D, E, F) subject to four constraints: $A + B = C$, $B + D = F$, $D + C = E$ and $A + F = E$. These constraints arise naturally from building a kind of tetrahedral arrangement of Hermitian matrices inside the vector space of Hermitian matrices according to Fig. 1. Consequently, throughout the paper, we refer to this generalization of Horn's problem as the *tetrahedral Horn's problem* and, by contrast, Horn's original problem, involving the sum of just two Hermitian matrices, as the *triangular Horn's problem*.

Beyond the superficial connection between the imposed constraints and the resulting shape formed in the vector space of Hermitian matrices (as in Fig. 1), in the special case of 2×2 Hermitian matrices, this connection is much deeper. It is well-known that the space of admissible eigenvalues for a triangular arrangement of 2×2 traceless¹ Hermitian matrices is in correspondence with the moduli space of Euclidean *triangles* in $\mathbb{R}^3$ [HK96; FKL10; Knu00]. As we shall see in Example 5, this correspondence generalizes in the sense that the admissible eigenvalues for a tetrahedral arrangements of 2×2 traceless Hermitian matrices is in correspondence with the moduli space of Euclidean *tetrahedra* in $\mathbb{R}^3$. While this latter result is straightforward to prove and also well-known, indeed it forms the basis of the semiclassical asymptotics of the $SU(2)$ Wigner $6j$ symbols [Rob99], it leads to a complete solution to the tetrahedral Horn's problem for 2×2 matrices. In Example 5, we will see that the moduli space of Euclidean tetrahedra in $\mathbb{R}^3$ is provided by the positivity of so-called Cayley-Menger determinant [SJS18].

Although we are ultimately interested in tackling the tetrahedral Horn's problem for $n \times n$ Hermitian matrices with $n > 2$, the $n = 2$ case already demonstrates a strong qualitative difference between the triangular and tetrahedral variants of Horn's problem. In particular, while the triangular Horn's problem is completely characterized by a *finite* number of *linear* inequalities in the eigenvalues (indeed the solution space is an instance of a moment polytope known as *Horn's polytope*), the solution space for the tetrahedral Horn's problem is in general *not a polytope*.

TODO: Finish the introduction

- Talk about connections between Horn's problem & Schubert calculus

¹One can always assume, without loss of generality, that the Hermitian matrices involved are traceless due to a simple shifting symmetry which is described in Section 3.

- talk about semiclassical asymptotics of $6j$ symbols
- talk about entropy cones and open problems regarding entropy cones
- talk about known submodular inequalities for PSD matrices
- describe the scope & results of the paper (more detailed abstract)
- describe structure of the paper

Tetrahedral eigenvalues & the tetrahedral Horn problem

prob:tri_problem

Problem 1 (Trigonal Horn problem). A triple $(\alpha, \beta, \gamma) \in (\mathbb{R}_{\text{dec}}^n)^{\times 3}$ of candidate eigenvalues are said to be *trigonal* if there exists a corresponding triple (A, B, C) of $d \times d$ Hermitian matrices satisfying (i) the linear constraint

$$A + B = C, \quad \text{(eq:tri_linear_constraints)} \quad (1)$$

and (ii) the three eigenvalue constraints

$$\text{eig}(A) = \alpha, \quad \text{eig}(B) = \beta, \quad \text{eig}(C) = \gamma. \quad \text{(eq:tri_eigenvalue_constraints)} \quad (2)$$

The problem of determining the set of trigonal eigenvalues (α, β, γ) will be referred to as the *trigonal Horn problem*.

prob:tet_problem

Problem 2 (Tetrahedral Horn problem). A sextuple $(\alpha, \beta, \gamma, \delta, \epsilon, \phi) \in (\mathbb{R}_{\text{dec}}^n)^{\times 6}$ of candidate eigenvalues are said to be *tetrahedral* if there exists a corresponding sextuple (A, B, C, D, E, F) of $n \times n$ Hermitian matrices satisfying (i) the four linear constraints

$$A + B = C, \quad B + D = F, \quad C + D = E, \quad A + F = E, \quad \text{(eq:tet_linear_constraints)} \quad (3)$$

and (ii) the six eigenvalue constraints

$$\text{eig}(A) = \alpha, \quad \text{eig}(B) = \beta, \quad \text{eig}(C) = \gamma, \quad \text{eig}(D) = \delta, \quad \text{eig}(E) = \epsilon, \quad \text{eig}(F) = \phi. \quad \text{(eq:tet_eigenvalue_constraints)} \quad (4)$$

The problem of determining the set of tetrahedral eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ will be referred to as the *tetrahedral Horn problem*.

exam:one_dim_soln

Example 3 (Tetrahedral Horn's problem for $n = 1$). Perhaps the simplest example of the tetrahedral Horn's problem occurs when $n = 1$; i.e., when Hermitian matrices and their eigenvalues are identified with real numbers. In this case, the tetrahedral Horn's problem reduces to the problem of which sextuples of real numbers (a, b, c, d, e, f) simultaneously satisfy

$$a + b = c, \quad b + d = f, \quad d + c = e, \quad a + f = e, \quad \text{(eq:one_dim_tet_conds)} \quad (5)$$

or equivalently, are such that $(a, b, c, d, e, f) \in \ker \Gamma$ where Γ is the 4×6 matrix of coefficients encoding the linear system from Eq. (5):

$$\Gamma := \begin{pmatrix} 1 & 1 & -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 1 & -1 & 0 \\ 1 & 0 & 0 & 0 & -1 & 1 \end{pmatrix}. \quad \text{(eq:tet_coeff_matrix)} \quad (6)$$

It can be shown that the above coefficient matrix has a three dimensional kernel and, in particular, one can parameterize the solution space by $(x, y, z) \in \mathbb{R}^3$ according to the formula

$$(a, b, c, d, e, f) = (x, y, x + y, z, x + y + z, y + z). \quad \text{(eq:one_dim_soln)} \quad (7)$$

Remark 4 (Traces are tetrahedral). In the original, trigonal version of Horn's problem, the eigenvalues (α, β, γ) of Hermitian matrices (A, B, C) satisfying $A + B = C$ necessarily must satisfy a basic equality condition known as the *trace condition*:

$$|\alpha| + |\beta| = |\gamma|, \quad \text{(eq:tri_trace_cond)} \quad (8)$$

where $|\lambda| := \sum_i \lambda_i$ is shorthand for the sum of the eigenvalues $\lambda = (\lambda_1 \geq \dots \geq \lambda_n)$. This condition must hold because $A + B = C$ implies $\text{Tr}(C) = \text{Tr}(A + B) = \text{Tr}(A) + \text{Tr}(B)$ and $\text{Tr}(X) = |\lambda|$ for any Hermitian matrix X with eigenvalues λ . For the tetrahedral Horn's problem, a set of tetrahedral eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ necessarily satisfy four *trace conditions*:

$$|\alpha| + |\beta| = |\gamma|, \quad |\beta| + |\delta| = |\phi|, \quad |\delta| + |\gamma| = |\epsilon|, \quad |\alpha| + |\phi| = |\epsilon|. \quad \text{(eq:tet_trace_conds)} \quad (9)$$

Comparing the tetrahedral trace conditions (Eq. (9)) with the $n = 1$ case of the tetrahedral Horn's problem (Eq. (3)), we see that a sextuple of eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ can only be tetrahedral if their traces, $(|\alpha|, |\beta|, |\gamma|, |\delta|, |\epsilon|, |\phi|)$, are tetrahedral.

Example 5 (Tetrahedral Horn's problem for $d = 2$). Here we articulate the solution to the tetrahedral Horn's problem for 2×2 Hermitian matrices. According to Corollary 13, one can restrict their focus to the case of *traceless* 2×2 Hermitian matrices without loss of generality. In this context, the sextuple of eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ can be identified with a sextuple of non-negative real numbers $(a, b, c, d, e, f) \in \mathbb{R}_{\geq 0}^6$ such that

$$\alpha = (a, -a), \quad \beta = (b, -b), \quad \gamma = (c, -c), \quad \delta = (d, -d), \quad \epsilon = (e, -e), \quad \phi = (f, -f). \quad (10)$$

Moreover, each traceless 2×2 Hermitian matrix, X , may be identified with a three-dimensional vector $(x_1, x_2, x_3) \in \mathbb{R}^3$ using the standard Pauli-matrix basis $\{\sigma_1, \sigma_2, \sigma_3\}$ according to the formula:

$$X = x_1\sigma_1 + x_2\sigma_2 + x_3\sigma_3 = \begin{pmatrix} x_3 & x_1 - ix_2 \\ x_1 + ix_2 & -x_3 \end{pmatrix}. \quad (11)$$

With respect to this parameterization, one sees that the eigenvalues of X are determined by the Euclidean length of the vector (x_1, x_2, x_3) :

$$\text{eig}(X) = \text{eig} \begin{pmatrix} x_3 & x_1 - ix_2 \\ x_1 + ix_2 & -x_3 \end{pmatrix} = (\|x\|, -\|x\|), \quad \text{where,} \quad \|x\| = \sqrt{x_1^2 + x_2^2 + x_3^2}. \quad (12)$$

Furthermore, if A is determined by the coordinates (a_1, a_2, a_3) and B by the coordinates (b_1, b_2, b_3) , then their sum $C = A + B$ is determined by the coordinates $(c_1, c_2, c_3) = (a_1 + b_1, a_2 + b_2, a_3 + b_3)$. Consequently, in this setting, the tetrahedral Horn's problem reduces to problem of describing the space of all possible edge-lengths (a, b, c, d, e, f) of a Euclidean tetrahedron in $\mathbb{R}^3$. This problem admits of a well-studied solution [Ric02; WD09], namely, the conjunction of a quadruplet of triangle-inequalities

$$|a - b| \leq c \leq a + b, \quad |b - d| \leq f \leq b + d, \quad |d - c| \leq e \leq d + c, \quad |a - f| \leq e \leq a + f, \quad (13)$$

(which ensure the possibility of forming a Euclidean triangle for each of the four faces), together with a single non-linear inequality of the form

$$D(a, b, c, d, e, f) = \frac{1}{288} \det \begin{pmatrix} 0 & 1 & 1 & 1 & 1 \\ 1 & 0 & a^2 & c^2 & e^2 \\ 1 & a^2 & 0 & b^2 & f^2 \\ 1 & c^2 & b^2 & 0 & d^2 \\ 1 & e^2 & f^2 & d^2 & 0 \end{pmatrix} \geq 0. \quad (14)$$

The quantity $D(a, b, c, d, e, f)$ is known as a Cayley-Menger determinant and, when $D(a, b, c, d, e, f)$ is positive and when the triangle inequalities above are also satisfied, $D(a, b, c, d, e, f)$ is always equal to *square* of the volume of the corresponding tetrahedron [SJS18, Equation 5.2]. Figure 2 depicts a three-dimensional projection of the solution space described by Eq. (13) and Eq. (14).

The tetrahedral Horn problem can be viewed as a generalization of the trigonal Horn problem in a variety of ways. First and foremost, the trigonal Horn problem is a special case of the tetrahedral Horn problem.

Proposition 6. *A triplet (α, β, γ) of eigenvalues is trigonal if and only if the sextuple $(\alpha, \beta, \gamma, 0, \gamma, \beta)$ is tetrahedral.*

Proof. If (α, β, γ) are the eigenvalues of (A, B, C) satisfying $A+B = C$, then one can see that $(A, B, C, 0, C, B)$ satisfy Eq. (3) and have eigenvalues $(\alpha, \beta, \gamma, 0, \gamma, \beta)$. Conversely, if $(\alpha, \beta, \gamma, 0, \gamma, \beta)$ are tetrahedral then it is straightforward to see that (α, β, γ) are trigonal. $\square$

More generally, it is possible to view the tetrahedral Horn problem as the conjunction of four separate instances of the trigonal Horn problem subject to additional matching constraints.

Lemma 7. *A sextuple of eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ is tetrahedral only if each of*

$$(\alpha, \beta, \gamma), \quad (\beta, \delta, \phi), \quad (\gamma, \delta, \epsilon), \quad (\alpha, \phi, \epsilon), \quad (15)$$

are trigonal.

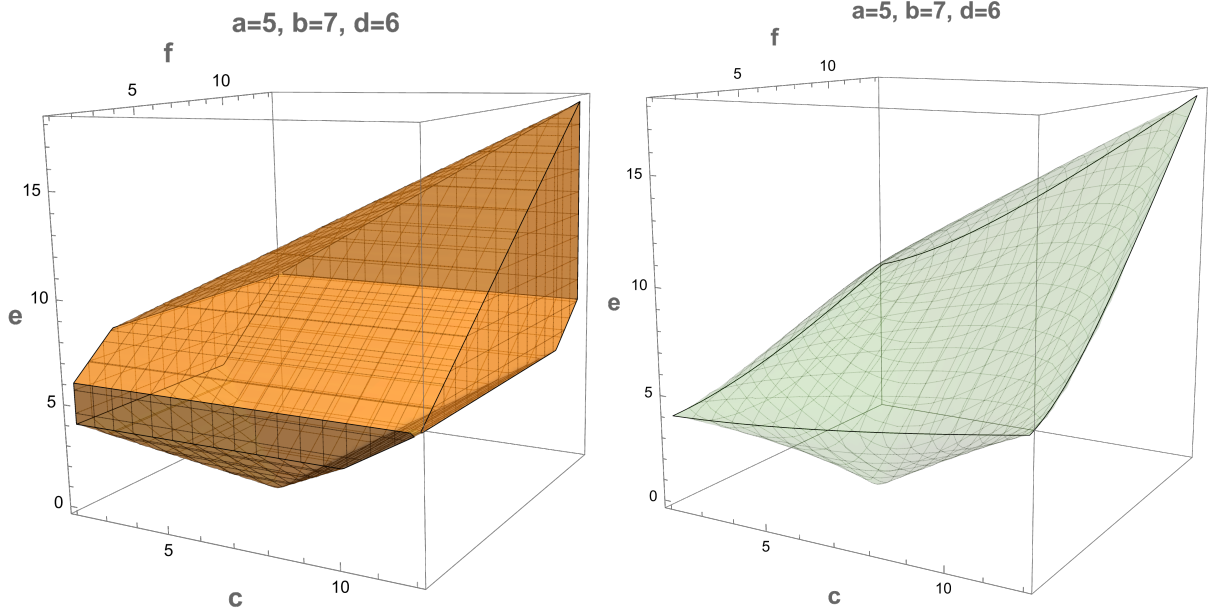


Figure 2: A slice of the set of admissible side-lengths (a, b, c, d, e, f) for Euclidean tetrahedron along the coordinates $(a, b, d) = (5, 7, 6)$ (Left) The polyhedron of values (c, e, f) satisfying the triangle inequalities for each face Eq. (13). (Right) The set of all values (c, e, f) satisfying both Eq. (13) and Eq. (14).

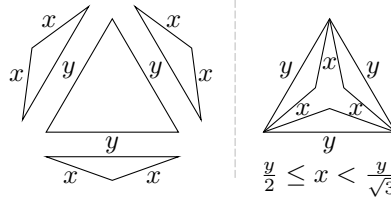


fig:frustrated-tet

Figure 3: A example of a sextuple of edge lengths, $(a, b, c, d, e, f) = (y, y, y, x, x, x)$, which satisfies all triangle inequality constraints whenever $y \leq 2x$, but cannot be the edge lengths of any Euclidean tetrahedron when $\sqrt{3}x < y$.

Proof. The claim follows from a relaxation of the conditions defining the tetrahedral Horn problem. In particular, there exists a sextuple (A, B, C, D, E, F) of Hermitian matrices with eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ satisfying Eq. (3) *only if* there exists twelve Hermitian matrices

$$(A, A', B, B', C, C', D, D', E, E', F, F') \tag{16}$$

with eigenvalues $(\alpha, \alpha, \beta, \beta, \gamma, \gamma, \delta, \delta, \epsilon, \epsilon, \phi, \phi)$ satisfying

$$A + B = C, \quad B' + D = F, \quad C' + D' = E, \quad A' + F' = E'. \tag{17}$$

Now notice that each of the four linear constraints above depend on a disjoint subset of the twelve matrices in Eq. (16) and therefore the respective eigenvalues are independently constrained. In other words, this relaxation of the tetrahedral Horn problem is equivalent to the statement that $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ are tetrahedral eigenvalues *only if* each of four triplets in Eq. (15) are trigonal. $\square$

3 Symmetries of tetrahedral eigenvalues

sec:symmetries

There are numerous types of symmetries present in the statement of Problem 2. The purpose of this section is to describe the scaling, inversion, tetrahedral, and shifting symmetries of the problem. These symmetries can, in many cases, be used to simplify the problem by selecting a canonical representative of the corresponding equivalence class; in other cases, it will be useful to leave Problem 2 as stated. Later we will comment on the possibility of additional hidden symmetries not previously mentioned.

Proposition 8 (Scaling symmetry). *Let $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ be a sextuple of candidate eigenvalues and let $s \in (0, \infty)$ be a positive real number. Then the scaled eigenvalues, $(s\alpha, s\beta, s\gamma, s\delta, s\epsilon, s\phi)$, are tetrahedral if and only if the original, unscaled eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ are tetrahedral.*

Proof. The statement follows from the fact that $\text{eig}(sX) = s \text{eig}(X)$ holds for all Hermitian matrices X and positive scalars s and that each of the constraints in Eq. (3) are homogeneous in the six Hermitian matrices (A, B, C, D, E, F) ; for example, we have $sA + sB - sC = 0$ if and only if $s(A + B - C) = 0$ if and only if $A + B - C = 0$. $\square$

One may wonder with or not it is possible to extend the the scaling symmetry to consider scaling by negative $s < 0$. Indeed, this can be accomplished, provided one accounts for the fact that negating a Hermitian matrices both negates and reverses the order of its eigenvalues.

prop:inversion

Proposition 9 (Inversion symmetry). *Let $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ be a sextuple of candidate eigenvalues. Given any list of non-increasing eigenvalues $\lambda = (\lambda_1 \geq \dots \geq \lambda_n)$, let λ^* denote the inverted list of resorted and negated eigenvalues: $\lambda^* = (-\lambda_d \geq \dots \geq -\lambda_1)$. Then $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ are tetrahedral if and only if $(\alpha^*, \beta^*, \gamma^*, \delta^*, \epsilon^*, \phi^*)$ are tetrahedral.*

Proof. The statement holds because (i) (A, B, C, D, E, F) satisfy Eq. (3) if and only if $(A', B', C', D', E', F') := (-A, -B, -C, -D, -E, -F)$ satisfy Eq. (3), and (ii) $\text{eig}(X) = \lambda$ if and only if $\text{eig}(-X) = \lambda^*$. $\square$

The inversion symmetry Proposition 9 is just one element belonging to a group of 48 symmetries consisting of inversions of eigenvalues, $\lambda \rightarrow \lambda^*$, or permutations of elements in the sextuple $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$.

prop:tet-symmetry

Proposition 10 (Oriented Tetrahedral symmetry). *The sextuple $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ of eigenvalues are tetrahedral if and only if any (and thus all) of the following sextuples of eigenvalues are tetrahedral:*

{eq:gens_of_tet_sym}

$$(\alpha \phi \epsilon \delta^* \gamma \beta) \quad (\gamma \epsilon^* \delta^* \phi \beta \alpha^*) \quad (18)$$

Moreover, these two operations generate a finite group of order 48 whose action, grouped by conjugacy classes, produces:

$$\begin{array}{cccc}
 (\alpha \ \beta \ \gamma \ \delta \ \epsilon \ \phi) & (\alpha^* \ \gamma \ \beta \ \delta \ \phi \ \epsilon) & (\alpha \ \gamma^* \ \beta^* \ \delta^* \ \phi^* \ \epsilon^*) & (\alpha^* \ \beta^* \ \gamma^* \ \delta^* \ \epsilon^* \ \phi^*) \\
 (\delta \ \beta \ \phi \ \alpha \ \epsilon \ \gamma) & (\beta^* \ \alpha^* \ \gamma^* \ \epsilon \ \delta \ \phi) & (\beta \ \alpha \ \gamma \ \epsilon^* \ \delta^* \ \phi^*) & (\delta^* \ \beta^* \ \phi^* \ \alpha^* \ \epsilon^* \ \gamma^*) \\
 (\delta^* \ \epsilon \ \gamma \ \alpha^* \ \beta \ \phi) & (\phi^* \ \beta \ \delta^* \ \gamma^* \ \epsilon^* \ \alpha^*) & (\phi \ \beta^* \ \delta \ \gamma \ \epsilon \ \alpha) & (\delta \ \epsilon^* \ \gamma^* \ \alpha \ \beta^* \ \phi^*) \\
 (\alpha \ \epsilon^* \ \phi^* \ \delta \ \beta^* \ \gamma^*) & (\epsilon \ \delta^* \ \gamma \ \beta^* \ \alpha \ \phi^*) & (\epsilon^* \ \delta \ \gamma^* \ \beta \ \alpha^* \ \phi) & (\alpha^* \ \epsilon \ \phi \ \delta^* \ \beta \ \gamma) \\
 (\beta \ \phi^* \ \delta^* \ \epsilon \ \gamma \ \alpha) & (\gamma \ \beta^* \ \alpha \ \phi \ \epsilon \ \delta) & (\gamma^* \ \beta \ \alpha^* \ \phi^* \ \epsilon^* \ \delta^*) & (\alpha^* \ \phi^* \ \epsilon^* \ \delta \ \gamma^* \ \beta^*) \\
 (\gamma \ \alpha^* \ \beta \ \phi^* \ \delta^* \ \epsilon^*) & (\alpha \ \phi \ \epsilon \ \delta^* \ \gamma \ \beta) & (\alpha^* \ \phi^* \ \epsilon^* \ \delta \ \gamma^* \ \beta^*) & (\beta^* \ \phi \ \delta \ \epsilon^* \ \gamma^* \ \alpha^*) \\
 (\epsilon \ \gamma^* \ \delta \ \beta \ \phi \ \alpha^*) & (\beta \ \delta \ \phi \ \epsilon^* \ \alpha^* \ \gamma^*) & (\beta^* \ \delta^* \ \phi^* \ \epsilon \ \alpha \ \gamma) & (\gamma^* \ \alpha \ \beta^* \ \phi \ \delta \ \epsilon) \\
 (\phi \ \alpha \ \epsilon \ \gamma^* \ \delta \ \beta^*) & (\delta \ \phi^* \ \beta^* \ \alpha^* \ \gamma^* \ \epsilon^*) & (\delta^* \ \phi \ \beta \ \alpha \ \gamma \ \epsilon) & (\epsilon^* \ \gamma \ \delta^* \ \beta^* \ \phi^* \ \alpha) \\
 (\phi^* \ \delta \ \beta^* \ \gamma \ \alpha \ \epsilon) & (\phi \ \epsilon^* \ \alpha^* \ \gamma \ \beta \ \delta^*) & (\phi^* \ \epsilon \ \alpha \ \gamma^* \ \beta^* \ \delta) & (\phi^* \ \alpha^* \ \epsilon^* \ \gamma \ \delta^* \ \beta) \\
 (\epsilon^* \ \phi \ \alpha^* \ \beta^* \ \gamma^* \ \delta) & (\epsilon^* \ \alpha \ \phi^* \ \beta \ \delta^* \ \gamma) & (\epsilon \ \alpha^* \ \phi \ \beta^* \ \delta \ \gamma^*) & (\phi \ \delta^* \ \beta \ \gamma^* \ \alpha^* \ \epsilon^*) \\
 (\gamma^* \ \delta^* \ \epsilon^* \ \phi \ \alpha^* \ \beta) & (\delta^* \ \gamma^* \ \epsilon^* \ \alpha \ \phi^* \ \beta^*) & (\delta \ \gamma \ \epsilon \ \alpha^* \ \phi \ \beta) & (\epsilon \ \phi^* \ \alpha \ \beta \ \gamma \ \delta^*) \\
 (\beta^* \ \gamma \ \alpha \ \epsilon^* \ \phi^* \ \delta^*) & (\gamma^* \ \epsilon \ \delta \ \phi^* \ \beta^* \ \alpha) & (\gamma \ \epsilon^* \ \delta^* \ \phi \ \beta \ \alpha^*) & (\gamma \ \delta \ \epsilon \ \phi^* \ \alpha \ \beta^*) \\
 & & & (\beta \ \gamma^* \ \alpha^* \ \epsilon \ \phi \ \delta)
 \end{array} \quad (19)$$

Proof. To begin, notice that each of the operations on $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ appearing in Eq. (18) are invertible (indeed the first operation has order two, while the second operation has order four). The claim that these two operations generate a finite group of order 48 can be checked explicitly using computer software. **TODO: Supply mathematica code used for this.** Now suppose $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ are the eigenvalues of a sextuple of Hermitian matrices (A, B, C, D, E, F) satisfying Eq. (3). Then, for example, the sextuple

$(A', B', C', D', E', F') = (A, F, E, -D, C, B)$ (which has the eigenvalues $(\alpha, \phi, \epsilon, \delta^*, \gamma, \beta)$) is also tetrahedral because

$$\begin{aligned} A' + B' - C' &= (A + F - E) = 0, \\ B' + D' - F' &= -(B + D - F) = 0, \\ C' + D' - E' &= -(C + D - E) = 0, \\ A' + F' - E' &= (A + B - C) = 0. \end{aligned} \tag{20}$$

Moreover, as the operation mapping $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ to $(\alpha^*, \gamma, \beta, \delta, \phi, \epsilon)$ is invertible, one also has that if $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ is not tetrahedral, then neither is $(\alpha^*, \gamma, \beta, \delta, \phi, \epsilon)$. An analogous argument holds for the operation mapping $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ to $(\gamma, \epsilon^*, \delta^*, \phi, \beta, \alpha^*)$ with (A', B', C', D', E', F') given by $(C, -E, -D, F, B, -A)$. $\square$

Remark 11. The group of order 48 appearing in Proposition 10 may be viewed as a subgroup of the group of all $6! \times 2^6 = 46,080$ possible signed permutations of 6 elements generated by permutations and negations of the elements of the sextuple $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$. Moreover, this subgroup is maximal in the sense that it is the maximal subgroup of permutations and negations of the columns of coefficient matrix Γ (appearing in Eq. (6)) which is equivalent to a permutation and negation of the rows of Γ (as in Eq. (20)). It can also be shown that this group of symmetries is isomorphic to the signed permutation group of three elements, i.e., $(\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2) \rtimes S_3$. In order to see this, it is helpful to arrange the sextuple $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ into a 2×3 grid using the following identification:

$$(\alpha \ \beta \ \gamma \ \delta \ \epsilon \ \phi) \cong \begin{pmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{pmatrix}. \tag{21}$$

Then one can see that each $\mathbb{Z}_2$ factor in the direct product $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$ acts faithfully on the sextuple by both negating a certain subset of the columns and permuting the upper and lower rows leaving just one column unchanged.

$$\begin{pmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{pmatrix} \leftrightarrow \begin{pmatrix} \alpha & \epsilon^* & \phi^* \\ \delta & \beta^* & \gamma^* \end{pmatrix}, \quad \begin{pmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{pmatrix} \leftrightarrow \begin{pmatrix} \delta & \beta & \phi \\ \alpha & \epsilon & \gamma \end{pmatrix}, \quad \begin{pmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{pmatrix} \leftrightarrow \begin{pmatrix} \delta^* & \epsilon & \gamma \\ \alpha^* & \beta & \phi \end{pmatrix}. \tag{22}$$

The symmetric group S_3 of order $3! = 6$ acts instead by permuting the columns and negating a subset of the entries in the top row, producing a subgroup of order 6 of the form:

$$\begin{pmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{pmatrix} \quad \begin{pmatrix} \beta & \gamma^* & \alpha^* \\ \epsilon & \phi & \delta \end{pmatrix} \quad \begin{pmatrix} \gamma^* & \alpha & \beta^* \\ \phi & \delta & \epsilon \end{pmatrix} \\ \begin{pmatrix} \alpha^* & \gamma & \beta \\ \delta & \phi & \epsilon \end{pmatrix} \quad \begin{pmatrix} \beta^* & \alpha^* & \gamma^* \\ \epsilon & \delta & \phi \end{pmatrix} \quad \begin{pmatrix} \gamma & \beta^* & \alpha \\ \phi & \epsilon & \delta \end{pmatrix} \tag{23}$$

Moreover, the above group of symmetries admits a finite presentation with respect to generators $g_1 \simeq (\alpha, \phi, \epsilon, \delta^*, \gamma, \beta)$ and $g_2 \simeq (\gamma, \epsilon^*, \delta^*, \phi, \beta, \alpha^*)$ as a free group over g_1, g_2 subject to the relations

$$\{g_1^2, g_2^4, (g_1 g_2)^6, (g_2 g_1 g_2 g_1 g_2)^2\}. \tag{24}$$

Proposition 12 (Shifting symmetry). *Let $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ be a sextuple of candidate eigenvalues and let $x, y, z \in \mathbb{R}$ be fixed. Given any list of non-increasing eigenvalues $\lambda = (\lambda_1 \geq \dots \geq \lambda_n)$, let $\lambda + x := (\lambda_1 + x \geq \dots \geq \lambda_d + x)$. Then define shifted eigenvalues by the formula*

$$\begin{aligned} \tilde{\alpha} &:= \alpha + x, & \tilde{\beta} &:= \beta + y, & \tilde{\gamma} &:= \gamma + x + y, \\ \tilde{\delta} &:= \delta + z, & \tilde{\epsilon} &:= \epsilon + x + y + z, & \tilde{\phi} &:= \phi + y + z. \end{aligned} \tag{25}$$

The shifted eigenvalues $(\tilde{\alpha}, \tilde{\beta}, \tilde{\gamma}, \tilde{\delta}, \tilde{\epsilon}, \tilde{\phi})$ are tetrahedral if and only if the unshifted eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ are tetrahedral.

Proof. Given a Hermitian matrix X with eigenvalues $\text{eig}(X) = \lambda$ and a constant $x \in \mathbb{R}$, the eigenvalues of $X + xI$ (where I is the identity matrix) are

$$\text{eig}(X + xI) = \lambda + x = (\lambda_1 + x \geq \dots \geq \lambda_d + x). \tag{26}$$

The statement then follows from Example 3 because

$$(A + xI, B + yI, C + (x + y)I, D + zI, E + (x + y + z)I, F + (y + z)I) \quad (27)$$

satisfy the tetrahedral constraints in Eq. (3) if and only if (A, B, C, D, E, F) satisfy the tetrahedral constraints in Eq. (3) $\square$

As a corollary of the shifting symmetries (Proposition 12) one concludes that to solve the tetrahedral Horn's problem, in a certain sense, it suffices to solve the tetrahedral Horn's problem merely for traceless matrices (Corollary 13) or, alternatively, for positive definite or positive semidefinite matrices (Corollary 14). cor:traceless

Corollary 13 (Traceless eigenvalues). *Let $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ be a sextuple of candidate eigenvalues and for any list of n eigenvalues $\lambda = (\lambda_1 \geq \dots \geq \lambda_n)$, let $\lambda^0 := \lambda - |\lambda|/n$ denote the traceless list of eigenvalues obtained by shifting λ by $-|\lambda|/n$. Then $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ are tetrahedral if and only if (i) their traces $(|\alpha|, |\beta|, |\gamma|, |\delta|, |\epsilon|, |\phi|)$ are tetrahedral (they satisfy Eq. (9)) and (ii) $(\alpha^0, \beta^0, \gamma^0, \delta^0, \epsilon^0, \phi^0)$ are tetrahedral.*

Proof. This “only if” portion of the statement follows from both the tetrahedral trace conditions (Eq. (9)) and applications of the shifting symmetry Proposition 12 with (x, y, z) values

$$x = -n^{-1} \text{Tr}(A), \quad y = -n^{-1} \text{Tr}(B) \quad z = -n^{-1} \text{Tr}(D). \quad (28)$$

The “if” portion of the statement follows from the shifting symmetry applied to the negated (x, y, z) values. $\square$

The next corollary of the shifting symmetry proves that to solve the tetrahedral Horn's problem it suffices to solve the tetrahedral Horn's problem for positive definite spectra only. Indeed, one can simply take the shifting parameters (x, y, z) appearing in Proposition 12 to be sufficiently large. The following corollary establishes, in a certain sense, the minimum shifting required to ensure strictly positive eigenvalues for all six Hermitian matrices (assuming some other basic inequalities hold). Note that in subsequent sections we will often explicitly assume the candidate eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ are either non-negative or positive for the sake of technical convenience or sometimes to establish a connection with density matrices from quantum theory (which are positive semidefinite matrices with unit trace). cor:psd

Corollary 14 (Positive eigenvalues). *If $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ are a tetrahedral sextuple of eigenvalues, then necessarily the smallest eigenvalues satisfy*

$$\gamma_d \geq \alpha_d + \beta_d, \quad \phi_d \geq \beta_d + \delta_d, \quad \epsilon_d \geq \delta_d + \gamma_d, \quad \epsilon_d \geq \alpha_d + \phi_d. \quad (29) \quad \text{\textcolor{blue}{eq:min_eig_ineqs}}$$

Moreover, if $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ are a sextuple of candidate eigenvalues satisfying Eq. (29), then for any $(x, y, z) \in \mathbb{R}^3$ satisfying eq:positive_shift

$$x > -\alpha_d, \quad y > -\beta_d, \quad z > -\delta_d, \quad (30)$$

the resulting shifted eigenvalues $(\tilde{\alpha}, \tilde{\beta}, \tilde{\gamma}, \tilde{\delta}, \tilde{\epsilon}, \tilde{\phi})$ (given by Eq. (25)) are all positive.

Proof. That tetrahedral eigenvalues satisfy Eq. (29) follows from four applications of the well-known inequality $\gamma_d \geq \alpha_d + \beta_d$ for the trigonal Horn's problem where $C = A + B$ [Bha01]. As a reminder, the smallest eigenvalue λ_n of a matrix X is characterized by a minimization: $\lambda_d = \min_{\|v\|=1} \langle v, Xv \rangle$ which produces

$$\gamma_d = \min_{\|v\|=1} \langle v, Cv \rangle = \min_{\|v\|=1} (\langle v, Av \rangle + \langle v, Bv \rangle) \geq \min_{\|v\|=1} \langle v, Av \rangle + \min_{\|w\|=1} \langle w, Bw \rangle = \alpha_d + \beta_d. \quad (31)$$

Four applications of this inequality produces Eq. (29). Now if (x, y, z) satisfy Eq. (30), we have for all j

$$\alpha_j + x \geq \alpha_d + x > 0, \quad (32)$$

meaning $\alpha + x = (\alpha_1 + x \geq \dots \geq \alpha_d + x)$ is strictly positive. Similar statements hold for $\beta, \gamma, \delta, \epsilon$ and ϕ ; for instance, for all j ,

$$\epsilon_j + x + y + z \geq \epsilon_d + x + y + z > \epsilon_d - \alpha_d - \beta_d - \delta_d \geq 0, \quad (33)$$

meaning $\epsilon + x + y + z$ is strictly positive also. $\square$

Remark 15. Although this section listed a handful of symmetries of the tetrahedral Horn's problem, there remains the possibility of additional symmetries which are unknown at this time. For example, in the case 2×2 Hermitian matrices, an additional class of symmetries on the set of edge-lengths (a, b, c, e, d, f) of tetrahedra, known as the *Regge symmetries*, hold [AI19; MS18; Boa07; PR68, Theorem 1.1]. If there exists a tetrahedron with edge lengths (a, b, c, d, e, f) , then there also exists a tetrahedron with edge lengths

$$(s_1 - a, s_1 - b, c, s_1 - d, s_1 - e, f) \quad \text{where} \quad s_1 = \frac{a + b + d + e}{2}. \quad (34)$$

Combining this symmetry with the permutational symmetries given by Proposition 10 (and noting that for the eigenvalues λ of traceless 2×2 Hermitian matrices, $\lambda = (x, -x)$, we have $\lambda^* = \lambda$) one concludes the co-existence of a family of six tetrahedra with edge lengths given by [VMK88, Chapter 9]:

$$\begin{aligned} &(a, b, c, d, e, f) && (s_1 - a, s_1 - b, c, s_1 - d, s_1 - e, f) \\ &(a, s_3 - b, s_3 - c, d, s_3 - e, s_3 - f) && (s_2 - a, b, s_2 - c, s_2 - d, e, s_2 - f) \\ &(s_2 - d, s_1 - e, s_3 - f, s_2 - a, s_1 - b, s_3 - c) && (s_1 - d, s_3 - e, s_2 - f, s_1 - a, s_3 - b, s_2 - c) \end{aligned} \quad (35)$$

where

$$s_1 = \frac{a + b + d + e}{2}, \quad s_2 = \frac{a + c + d + f}{2}, \quad s_3 = \frac{b + c + e + f}{2}. \quad (36)$$

Whether there exists analogous types of symmetries for the tetrahedral Horn's problem when $d > 2$ remains an open problem. **TODO:** Describe how Regge symmetries emerge from $SU(2)$ $6j$ symbol symmetries [Boa07], how they can be viewed as a scissors-congruence of Euclidean tetrahedra perserving both the volume invariants and Dhen invariants [AI19], and how higher rank Regge symmetries are connected to the eigenvalue conjecture [MS18].

4 Eigenvalues of blocks of positive semidefinite matrices

[sec:blocks_of_psd_matrices](#)

The purpose of this section is to reformulate both the trigonal and tetrahedral Horn's problem as a instance of a slightly more general kind of eigenvalue problem involving positive semidefinite matrices. Throughout this section, we will restrict our focus to the case of positive semidefinite matrices and their eigenvalues. As a reminder we note that this can be done without loss of generality due to Corollary 14.

Given a positive semidefinite matrix, X , with eigenvalues $\text{eig}(X) = \chi = (\chi_1 \geq \dots \geq \chi_n \geq 0)$ and eigendecomposition $X = \sum_{i=1}^n \chi_i \Pi_i$, one can define $X^{\frac{1}{2}}$ by $X^{\frac{1}{2}} := \sum_{i=1}^n \chi_i^{\frac{1}{2}} \Pi_i$. Inspired by [Bha97, Theorem IV.2.13], we obtain the following fundamental lemma relating the eigenvalues γ of a sum $C = A + B$ of $n \times n$ Hermitian matrices to the eigenvalues of a larger $2n \times 2n$ matrix M built from A and B .

[lem:trigonal_block](#)

Lemma 16. *Let (A, B, C) be a triple of $n \times n$ Hermitian matrices with eigenvalues (α, β, γ) such that $C = A + B$. Let $\Sigma_{A,B}$ be the $2n \times 2n$ Hermitian matrix*

$$\Sigma_{A,B} = \begin{bmatrix} A & A^{\frac{1}{2}} B^{\frac{1}{2}} \\ B^{\frac{1}{2}} A^{\frac{1}{2}} & B \end{bmatrix}. \quad (37)$$

Then $\Sigma_{A,B} \geq 0$ is positive semidefinite, has rank at most n and has eigenvalues given by γ

$$\text{eig}(\Sigma_{A,B}) \simeq \text{eig}(A + B) = \text{eig}(C) = \gamma. \quad (38)$$

Proof. That $\Sigma_{A,B}$ is positive semidefinite and has rank at most n follows because $\Sigma_{A,B}$ can be decomposed as $\Sigma_{A,B} = T^* T$ where T is the $n \times 2n$ block matrix

$$T = \begin{bmatrix} A^{\frac{1}{2}} & B^{\frac{1}{2}} \end{bmatrix}. \quad (39)$$

That $\Sigma_{A,B}$ and $A + B$ share the same eigenvalues (up to padding by n zeros) follows because for any rectangular matrix T , the eigenvalues of $T^* T$ and $T T^*$ are equal (up to padding by zeros) and in this instance

$$T^* T = \begin{bmatrix} A^{\frac{1}{2}} \\ B^{\frac{1}{2}} \end{bmatrix} \begin{bmatrix} A^{\frac{1}{2}} & B^{\frac{1}{2}} \end{bmatrix} = \begin{bmatrix} A & A^{\frac{1}{2}} B^{\frac{1}{2}} \\ B^{\frac{1}{2}} A^{\frac{1}{2}} & B \end{bmatrix} = \Sigma_{A,B} \quad T T^* = \begin{bmatrix} A^{\frac{1}{2}} & B^{\frac{1}{2}} \end{bmatrix} \begin{bmatrix} A^{\frac{1}{2}} \\ B^{\frac{1}{2}} \end{bmatrix} = A + B. \quad (40)$$

□

From the perspective of Lemma 16, one can view the trigonal Horn problem involving the eigenvalues (α, β, γ) of $n \times n$ Hermitian matrices (A, B, C) satisfying $A + B = C$ as a special case of a more general eigenvalue problem concerned with the relations between the eigenvalues of $2n \times 2n$ Hermitian matrices Σ of the form

$$\Sigma = \begin{bmatrix} A & X \\ X^* & B \end{bmatrix}, \quad (41) \quad \text{\texttt{eq:block_ABX}}$$

and the eigenvalues of its two $n \times n$ diagonal blocks A and B . It is known, by [HJ12, Theorem 7.7.9] or more generally [MKX19, Theorem 5.11] that if Σ is a 2×2 block matrix such that $A, B \geq 0$, then $\Sigma \geq 0$ if and only if the off-diagonal block X is of the form

$$X = A^{\frac{1}{2}} K B^{\frac{1}{2}} \quad (42)$$

for some contraction matrix K (largest eigenvalue of $(K^* K)^{\frac{1}{2}}$ is at most unity). Additionally, when $X = A^{\frac{1}{2}} K B^{\frac{1}{2}}$, one has the decomposition

$$\Sigma = \begin{bmatrix} A & X \\ X^* & B \end{bmatrix} = \begin{bmatrix} A^{\frac{1}{2}} & 0 \\ 0 & B^{\frac{1}{2}} \end{bmatrix} \begin{bmatrix} I & K \\ K^* & I \end{bmatrix} \begin{bmatrix} A^{\frac{1}{2}} & 0 \\ 0 & B^{\frac{1}{2}} \end{bmatrix}, \quad (43)$$

noting that K is a contraction if and only if the middle block matrix above is positive semidefinite [MKX19, Lemma 5.10].

Therefore, the special case covered by Lemma 16 corresponds to the case where the corresponding contraction matrix K is the $n \times n$ identity matrix $K = I$ (or more generally any contractive matrix satisfying

$A^{\frac{1}{2}}KB^{\frac{1}{2}} = A^{\frac{1}{2}}B^{\frac{1}{2}}$). A simple observation which distinguishes between eigenvalue problem concerning diagonal blocks and the eigenvalue problem concerning sums is simply that the rank of Σ can in general be up to $2n$ whereas the rank of the sum $C = A + B$ is at most n . Despite this additional freedom, there remains non-trivial relations constraining the eigenvalues of Σ and $A + B$ in terms of properties of X (see [Gum+18; BL22]). Nevertheless, when the rank of Σ is constrained to be at most n , we obtain the following theorem which provides an equivalent formulation of the trigonal Horn problem. thm:trigonal_block

Theorem 17. *Let (α, β, γ) be a triple of positive semidefinite eigenvalues of length at most n . The following are equivalent:*

- (i) *there exists a triple of Hermitian matrices (A, B, C) with eigenvalues (α, β, γ) satisfying $A + B = C$,*
- (ii) *there exists an $2n \times 2n$ Hermitian matrix,*

$$\Sigma = \begin{bmatrix} A' & X \\ X^* & B' \end{bmatrix}, \quad (44)$$

such that $\text{eig}(A') = \alpha$, $\text{eig}(B') = \beta$, and $\text{eig}(\Sigma) = \gamma$.

Proof. Since we explicitly assume all of the eigenvalues in the triple (α, β, γ) are non-negative, the Hermitian matrices A, B, C , and Σ, A', B' involved in the statements above are necessarily positive semidefinite. The claim (i) $\implies$ (ii) follows from Lemma 16 by setting $(\Sigma, A', B') = (\Sigma_{A,B}, A, B)$. That (ii) $\implies$ (i) follows by noticing that if Σ has eigenvalues γ , then it has rank at most n . Therefore, as $\Sigma \geq 0$, there exists a $n \times 2n$ block matrix T of the form

$$T = \begin{bmatrix} T_A & T_B \end{bmatrix} \quad (45)$$

such that $\Sigma = T^*T$. The diagonal blocks of Σ are therefore $A' = T_A^*T_A$, and $B' = T_B^*T_B$. As in the proof of Lemma 16, we note that T^*T and TT^* have the same eigenvalues up to padding by zeros. Therefore, by letting $A = T_A T_A^*$ and $B = T_B T_B^*$ we see that

$$\begin{aligned} \text{eig}(A) &= \text{eig}(T_A T_A^*) \simeq \text{eig}(T_A^* T_A) = \text{eig}(A') = \alpha, \\ \text{eig}(B) &= \text{eig}(T_B T_B^*) \simeq \text{eig}(T_B^* T_B) = \text{eig}(B') = \beta, \end{aligned} \quad (46)$$

while the eigenvalues of $C = A + B = T_A T_A^* + T_B T_B^*$ match the eigenvalues of γ because

$$\text{eig}(C) = \text{eig}(A + B) = \text{eig}(T_A T_A^* + T_B T_B^*) = \text{eig}(TT^*) \simeq \text{eig}(T^*T) = \text{eig}(\Sigma) = \gamma. \quad (47)$$

□

As a corollary of this correspondence between the trigonal Horn problem and the eigenvalues of blocks of 2×2 matrices, we obtain an equivalent formulation of the tetrahedral Horn problem. cor:tetrahedral_block

Corollary 18. *Let $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ be a sextuple of positive semidefinite eigenvalues of length at most n . Then $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ are tetrahedral if and only if there exists a $3n \times 3n$ matrix Σ of the form*

$$\Sigma = \begin{bmatrix} A' & Z & Y \\ Z^* & B' & X \\ Y^* & X^* & D' \end{bmatrix}, \quad (48) \quad \text{eq:block_3by3}$$

such that

$$\text{eig}(A') = \alpha, \quad \text{eig}(B') = \beta, \quad \text{eig}\begin{pmatrix} A' & Z \\ Z^* & B' \end{pmatrix} = \gamma, \quad \text{eig}(D') = \delta, \quad \text{eig}(\Sigma) = \epsilon, \quad \text{eig}\begin{pmatrix} B' & X \\ X^* & D' \end{pmatrix} = \phi. \quad (49) \quad \text{eq:block_eigenvalue_tet}$$

Proof. The proof is analogous to the proofs of Lemma 16 and Theorem 17. First we prove the “only if” portion of the claim which is a corollary of Lemma 16. Suppose (A, B, C, D, E, F) is a tetrahedral sextuple of Hermitian matrices with eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \gamma)$. Let $\Sigma = \Sigma_{A,B,D}$ be given by

$$\Sigma_{A,B,D} = \begin{bmatrix} A^{\frac{1}{2}} & B^{\frac{1}{2}} & D^{\frac{1}{2}} \end{bmatrix}^* \begin{bmatrix} A^{\frac{1}{2}} & B^{\frac{1}{2}} & D^{\frac{1}{2}} \end{bmatrix} = \begin{bmatrix} A & A^{\frac{1}{2}}B^{\frac{1}{2}} & A^{\frac{1}{2}}D^{\frac{1}{2}} \\ B^{\frac{1}{2}}A^{\frac{1}{2}} & B & B^{\frac{1}{2}}D^{\frac{1}{2}} \\ D^{\frac{1}{2}}A^{\frac{1}{2}} & D^{\frac{1}{2}}B^{\frac{1}{2}} & D \end{bmatrix} \quad (50)$$

Then the eigenvalues of the corresponding principal submatrices of Σ satisfy Eq. (49) by Lemma 16. Now we prove the “if” portion of the claim which is a corollary of Theorem 17. If Σ is a matrix of form given by Eq. (48) with eigenvalues ϵ with length at most n , then Σ is positive semidefinite with rank at most n . Therefore, there exists an $n \times 3n$ block matrix

$$T = \begin{bmatrix} T_A & T_B & T_D \end{bmatrix} \quad (51)$$

such that $\Sigma = T^*T$ and thus

$$\begin{aligned} A' &= T_A^*T_A, & B' &= T_B^*T_B, & D' &= T_D^*T_D, \\ X &= T_B^*T_D, & Y &= T_A^*T_D, & Z &= T_A^*T_B. \end{aligned} \quad (52)$$

Now let (A, B, C, D, E, F) be given by

$$\begin{aligned} A &= T_A T_A^*, & B &= T_B T_B^*, & C &= T_A T_A^* + T_B T_B^*, \\ D &= T_D T_D^*, & E &= T_A T_A^* + T_B T_B^* + T_D T_D^*, & F &= T_B T_B^* + T_D T_D^*. \end{aligned} \quad (53)$$

From here, one may directly verify that these $n \times n$ Hermitian matrices satisfy Eq. (3) and have eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ (using a similar argument presented in the proof of Theorem 17); i.e., one concludes that $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ are tetrahedral. $\square$

Remark 19. The correspondence between the tetrahedral Horn problem and the eigenvalues of principal submatrices of 3×3 matrices provided by Corollary 18 enables us to import a myriad of known results concerning the eigenvalues of principal submatrices. Specifically, the results due to Audenaert, Hiai, and Petz [AHP10], Lami et al. [Lam+17], Niculescu [Nic25], Paksoy, Turkmen, and Zhang [PTZ14], Lewin and Sabin [LS14]. There is a wealth of results relating to inequalities of determinants of blocks of PSD matrices, including Hadamard, Fischer, Kotljanskii, and Szasz [JB93].

Many of the results concerning the eigenvalues of principal submatrices are born out of the correspondence between PSD matrices and covariance matrices of Gaussian multivariate random variables [Lam+17]. From this perspective, the Gaussian entropy region has been partially characterized by Shadbakht and Hassibi [SH11; HS08]. From this perspective, one may view Horn problems are kinds of “spectral” classical marginal problems for blocks of Gaussian random variables.

5 Inequalities for tetrahedral eigenvalues

- In [AZ99], Ando and Zhan proved a subadditivity inequality for non-negative operator monotone functions $f(t)$ on $[0, \infty)$. Specifically, for any PSD $A, B \geq 0$, and an unitarily invariant norm,

$$\|f(A) + f(B)\| \geq \|f(A + B)\|. \quad (54)$$

- Later in [Kos06], Kosem considered a generalization of the Ando-Zhan theorem which asks if for every non-negative operator monotone function $f(t)$ and every unitarily invariant norm and $A, B, C \geq 0$ PSD whether

$$\|f(A + B) + f(B + C) - f(B)\| \geq \|f(A + B + C)\|? \quad (55)$$

Kosem showed that this statement fails to hold for 2×2 PSD matrices when $f(t) = \frac{t}{t+1}$ and the norm $\|\cdot\|$ is the Ky Fan 2-norm (which for PSD matrices $X = (X^\dagger X)^{\frac{1}{2}}$ has the form $\|X\|_{(k)} = \sum_{i=1}^k \text{eig}_i(X)$).

- See also the inequalities presented by Aujla and Bourin [AB07], namely that if f is monotone concave on $[0, \infty)$ then there exists unitaries U, V such that as operators

$$f(A + B) \leq Uf(A)U^* + Vf(B)V^*. \quad (56)$$

- In [AHP10], Audenaert, Hiai, and Petz investigate the set of so-called *strongly subadditive functions* $f : \mathbb{R}^+ \rightarrow \mathbb{R}$ such that

$$\text{Tr } f(P_{123}XP_{123}) + \text{Tr } f(P_2XP_2) \leq \text{Tr } f(P_{12}XP_{12}) + \text{Tr } f(P_{23}XP_{23}), \quad (57)$$

holds for all positive operators X and orthogonal projection operators P_1, P_2, P_3 where $P_{12} = P_1 + P_2$, $P_{23} = P_2 + P_3$ and $P_{123} = P_1 + P_2 + P_3$.

- It seems to be that analogous results were also derived by Friedland and Gaubert [FG13].
- Later, in [Lam+17], Lami et al. is concerned with the option $f(x) = \ln x$ which pertains to log-determinant inequalities and its connection to entropy inequalities.
- It seems probable that could be a unifying submodularity inequality (maybe related to [GJL17]?) prop:simple_quadratic

Proposition 20. *Let $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ be positive semidefinite tetrahedral eigenvalues. Then*

$$\sum_j \alpha_j \delta_{d-j+1} \leq \frac{1}{2} \left(\sum_j \beta_j^2 + \sum_j \epsilon_j^2 - \sum_j \gamma_j^2 - \sum_j \phi_j^2 \right) \leq \sum_j \alpha_j \delta_j \quad (58)$$

Proof. Let (A, B, C, D, E, F) be positive semidefinite matrices with eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ satisfying $A + B = C$, $B + D = F$, and $A + F = C + D = E$. The following identity holds:

$$\begin{aligned} & \text{Tr}[B^2] + \text{Tr}[E^2] - \text{Tr}[C^2] - \text{Tr}[F^2] \\ &= \text{Tr}[(B)^2] + \text{Tr}[(A + B + D)^2] - \text{Tr}[(A + B)^2] - \text{Tr}[(B + D)^2] \\ &= (\text{Tr}[B^2]) + (\text{Tr}[A^2] + \text{Tr}[B^2] + \text{Tr}[D^2] + 2 \text{Tr}[AB] + 2 \text{Tr}[BD] + 2 \text{Tr}[AD]) \\ &\quad - (\text{Tr}[A^2] + \text{Tr}[B^2] + 2 \text{Tr}[AB]) - (\text{Tr}[B^2] + \text{Tr}[D^2] + 2 \text{Tr}[BD]) \\ &= 2 \text{Tr}[AD]. \end{aligned} \quad (59)$$

While the value of $\text{Tr}(AD)$ is not determined the eigenvalues $\alpha = \text{eig}(A)$ and $\delta = \text{eig}(D)$, one has the following bounds [MOA10, H.1.g & H.1.h]:

$$\sum_j \alpha_j \delta_{d-j+1} \leq \text{Tr}[AD] \leq \sum_j \alpha_j \delta_j, \quad (60)$$

where $\delta_{d-j+1} = \delta_j^*$. The claim then follows because $\text{Tr}[X^2] = \sum_j \lambda_j^2$ for any Hermitian operator X with eigenvalues $\text{eig}(X) = \lambda$. $\square$

5.1 Trigonal Horn inequalities

sec:horn_ineqs

There are a wealth of inequalities constraining the set of tetrahedral eigenvalues which may be viewed as generalization of the triangle inequalities which hold for case where $d = 2$ (see Eq. (13)). From Lemma 7 we see that $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ are tetrahedral only if each of (α, β, γ) , (β, γ, ϕ) , $(\gamma, \delta, \epsilon)$, and (α, ϕ, ϵ) are trigonal. Consequently, every inequality constraining trigonal eigenvalues can be lifted to four inequalities constraining tetrahedral eigenvalues. The purpose of this section, therefore, is to briefly summarize some known inequalities which hold for triangular eigenvalues (α, β, γ) . Of particular interest are Horn's inequalities [Hor62] which are known to be necessary and sufficient [Kly98; KT99]. For a more complete historical account, the reader is directed to the survey articles [Bha01; Ful98].

In addition to the aforementioned trace condition, $\sum_i \alpha_i + \sum_j \beta_j = \sum_k \gamma_k$, one always has the following pair of elementary inequalities,

$$\alpha_1 + \beta_1 \geq \gamma_1, \quad \text{and} \quad \alpha_d + \beta_d \leq \gamma_d, \quad (61)$$

which constrain the largest and smallest eigenvalues respectively. The first somewhat non-trivial inequality, derived by Weyl in 1912, is now known as *Weyl's inequality* and states [Wey12]

$$\alpha_i + \beta_j \geq \gamma_{i+j-1}, \quad (62)$$

provided that $i + j - 1 \leq d$. For 2×2 matrices ($d = 2$) one can see that these necessary inequalities are also sufficient to ensure that (α, β, γ) are trigonal.²

In 1949, Fan derived an *majorization* inequality constraining the first r largest eigenvalues [Fan49]:

$$\sum_{i=1}^r \alpha_i + \sum_{j=1}^r \beta_j \geq \sum_{k=1}^r \gamma_k, \quad (63)$$

and shortly thereafter Lidskii & Wielandt found proofs of another family of inequalities which hold for any non-empty subset $I = \{i_1, \dots, i_r\} \subseteq \{1, \dots, d\}$ of indices of size r of the form

$$\sum_{j=1}^r \alpha_{i_j} + \sum_{j=1}^r \beta_j \geq \sum_{j=1}^r \gamma_{i_j}, \quad (64)$$

along with the inequality obtained by exchanging α and β . Finally, in 1962, Horn proved that these inequalities together with

$$\alpha_1 + \alpha_3 + \beta_1 + \beta_3 \geq \gamma_2 + \gamma_3 \quad (65)$$

are sufficient to ensure that (α, β, γ) are trigonal for the case where $d = 3$ [Hor62]. Moreover, also in [Hor62], Horn conjectured that in addition to the trace condition, a set of inequalities of the form

$$\sum_{i \in I} \alpha_i + \sum_{j \in J} \beta_j \geq \sum_{k \in K} \gamma_k \quad (66)$$

(for some inductively defined subsets $I, J, K \subset \{1, \dots, d\}$ of equal cardinality) was both necessary and sufficient to ensure that (α, β, γ) are trigonal.

TODO: Articulate how the four sets of Horn inequalities are necessary but insufficient conditions for the tetrahedral Horn problem. **TODO:** Articulate how the Horn inequalities can be inductively defined but also defined in terms of non-vanishing LR coefficients.

5.2 Submodular inequalities

DISCLAIMER: In this section I've switched to the A, B, C convention instead of the A, B, D convention. When discussing the tetrahedral Horn problem and its relation to the ordinary Horn problem or when matching the problem to the properties of the $6j$ symbols, the A, B, D convention is preferred. However when talking about submodular/strong subadditivity inequalities, the A, B, C convention is preferred. Before completing the paper, it will be desirable to establish a globally consistent convention.

Throughout this section, we will be in inequalities constraining the eigenvalues of $B, A + B, B + C$, and $A + B + C$ for PSD matrices $A, B, C \geq 0$. Note that under this formulation, we are not interested in the

²After shifting to traceless eigenvalues using Corollary 13, $\alpha = (a, -a)$, $\beta = (b, -b)$ and $\gamma = (c, -c)$, these inequalities become equivalent to the triangle inequalities $c \leq a + b$ and $c \geq \max\{b - a, a - b\}$.

eigenvalues of either A or C . An equivalent formulation is to consider eigenvalues of four PSD matrices W, X, Y, Z such that $X + Y = W + Z$ and $W \geq X \geq Z$ and $W \geq Y \geq Z$. That these two formulations are the same follows from the identification $X = B + C$, $Y = A + B$, $Z = B$ and $W = A + B + C$.

Bouhtou, Gaubert, and Sagnol claim to prove the following family of inequalities [BGS08, Lemma 5.3].
lem:bouhtou_ineq

Lemma 21. *Let $A, B, C \geq 0$ be positive semidefinite matrices and let $p \in [0, 1]$. Then*

$$\sum_j \lambda_j(A + B)^p + \sum_j \lambda_j(B + C)^p \geq \sum_j \lambda_j(B)^p + \sum_j \lambda_j(A + B + C)^p. \quad (67)$$

The following operator inequality can be used to prove a number of inequalities [TCL11, Lemma 2.2].
lem:mme_operator_ineq

Lemma 22. *Let $A, B, C \geq 0$ be positive semidefinite matrices and let k be a positive integer. Then*

$$(A + B + C)^{\otimes k} + B^{\otimes k} \geq (A + B)^{\otimes k} + (B + C)^{\otimes k}. \quad (68)$$

Proof. The result is obtained from applying a non-commutative multinomial expansion and noting that every summand on the right-hand-side above also appears on the left-hand-side above. $\square$

To obtain an inequality which depends only on the eigenvalues of $B, A + B, B + C$, and $A + B + C$, one can simply pair the positive semidefinite operator in Lemma 22 with an $SU(n)$ -equivariant and positive semidefinite operator:
cor:isotypic_ineq

Corollary 23. *Let $A, B, C \geq 0$ be positive semidefinite matrices and let $\lambda \vdash k$ be a Young diagram with k boxes. Then*

$$\text{Tr}[P_\lambda(A + B + C)^{\otimes k}] + \text{Tr}[P_\lambda B^{\otimes k}] \geq \text{Tr}[P_\lambda(A + B)^{\otimes k}] + \text{Tr}[P_\lambda(B + C)^{\otimes k}]. \quad (69)$$

Moreover, as $\text{Tr}[P_\lambda X^{\otimes k}] \propto E_\lambda(X)^3$, where $E_\lambda(X) = s_\lambda(\text{eig}(X))$ is the Schur polynomial of the eigenvalues of X , we equivalently have

$$s_\lambda(A + B + C) + s_\lambda(B) \geq s_\lambda(A + B) + s_\lambda(B + C). \quad (70)$$

As a special case of the previous corollary, we obtain an inequality which complements Proposition 20.
lem:quadratic_ineq_dual

Corollary 24. *Let $A, B, C \geq 0$ be positive semidefinite matrices, then*

$$\text{Tr}[A + B]^2 - \text{Tr}[(A + B)^2] + \text{Tr}[B + C]^2 - \text{Tr}[(B + C)^2] \leq \text{Tr}[B]^2 - \text{Tr}[B^2] + \text{Tr}[A + B + C]^2 - \text{Tr}[(A + B + C)^2]. \quad (71)$$

Proof. This result is a special case of Corollary 23 below when $\lambda = (1, 1)$ because

$$\text{Tr}[P_{(1,1)} X^{\otimes 2}] = \frac{1}{2} (\text{Tr}(X)^2 - \text{Tr}(X^2)). \quad (72)$$

$\square$

Paksoy, Turkmen, and Zhang proved a family of polynomial inequalities which are generally not functions of just the eigenvalues [PTZ14]. These inequalities are in terms a so-called generalized matrix function first defined by Marcus and Minc [MM65].

Definition 25 (Generalized matrix function). Let H be a subgroup of S_n and let χ be an irreducible character of H . The *generalized matrix function* for the character χ of H is

$$d_\chi^H(X) = \sum_{\sigma \in H} \chi(\sigma) \prod_i X_{i, \sigma(i)} \quad (73)$$

The notion of a *generalized matrix function* generalizes the notion of an *immanant* (when $H = S_n$) which, in turn, contains as special cases the notion of a permanent and determinant [HM21].

³Specifically the proportionality constant here is equal to $\dim W_\lambda$ where W_λ is the irreducible representation of S_k labelled by λ .

Theorem 26. Let $A, B, C \geq 0$ be positive semidefinite $n \times n$ matrices and let H be a subgroup of S_n with irreducible character χ . Then

$$d_\chi^H(A + B + C) + d_\chi^H(B) \geq d_\chi^H(A + B) + d_\chi^H(B + C). \quad (74)$$

Proof. See [PTZ14]. □

As a corollary of Theorem 26, we obtain the following determinant and permanent inequalities.

Corollary 27. Let $A, B, C \geq 0$ be positive semidefinite matrices. Then

$$\det(A + B + C) + \det(B) \geq \det(A + B) + \det(B + C), \quad (75) \quad \text{\texttt{feq:additive_det}}$$

$$\text{per}(A + B + C) + \text{per}(B) \geq \text{per}(A + B) + \text{per}(B + C). \quad (76)$$

Note that with the exception of the above determinant inequality (Eq. (75)), the inequalities provided by Theorem 26 are generally not equivalent to inequalities involving only eigenvalues.

The linear determinant inequality above can be seen as an extension of the well-known inequality $\det(A + B) \geq \det(A) + \det(B)$. One can make Eq. (75) symmetric in A, B, C by adding together three copies of Eq. (75) to obtain

$$3 \det(A + B + C) + \det(A) + \det(B) + \det(C) \geq 2 \det(A + B) + 2 \det(B + C) + 2 \det(A + C). \quad (77)$$

However, this inequality evidently no more constraining that the conjunction of the three copies of Eq. (75) from which it is produced. Amazingly, Lin has shown that one also has a Hlawka-like inequality [Lin14]:

$$\det(A + B + C) + \det(A) + \det(B) + \det(C) \geq \det(A + B) + \det(B + C) + \det(A + C). \quad (78) \quad \text{\texttt{feq:lin_hlawka_ineq}}$$

The Hlawka-like inequalities also have associated operator inequalities [BS15]:

$$(A + B + C)^{\otimes k} + A^{\otimes k} + B^{\otimes k} + C^{\otimes k} \geq (A + B)^{\otimes k} + (B + C)^{\otimes k} + (A + C)^{\otimes k}. \quad (79)$$

More recently Niculescu and Sra have shown a myriad of related inequalities and generalizations [NS23]. For instance

$$e_k(A) + e_k(B) + e_k(C) + e_k(A + B + C) \geq e_k(A + B) + e_k(B + C) + e_k(A + C), \quad (80)$$

as well as reporting other inequalities [NS23, Section 6.2] by other authors:

$$\det^{\frac{1}{2}} A + \det^{\frac{1}{2}} B + \det^{\frac{1}{2}} C + \det^{\frac{1}{2}}(A + B + C) \leq \det^{\frac{1}{2}}(A + B) + \det^{\frac{1}{2}}(B + C) + \det^{\frac{1}{2}}(A + C), \quad (81)$$

$$\det^{\frac{1}{n}}(A + B) \det^{\frac{1}{n}}(A + C) \geq \det^{\frac{1}{n}}(B) \det^{\frac{1}{n}}(C) + \det^{\frac{1}{n}}(A + B + C) \det^{\frac{1}{n}}(A) \quad (82)$$

TC: Confirm the above two inequalities hold for complex Hermitian matrices, not just real symmetric ones.

One also has another determinant inequality which is non-linear in determinants [TCL11, Lemma 2.3]. \texttt{lem:det_prod_ineq}

Lemma 28. Let $A, B, C \geq 0$ be positive semidefinite matrices. Then

$$\det(A + B + C) \det(B) \leq \det(A + B) \det(B + C). \quad (83)$$

Note that Lemma 28 can be seen as a special of Lemma 21 in the limit as $p \rightarrow 0$; the derivative of Lemma 21 in p at $p = 0$ yields

$$\sum_j \log \lambda_j(A + B) + \sum_j \log \lambda_j(B + C) \geq \sum_j \log \lambda_j(B) + \sum_j \log \lambda_j(A + B + C), \quad (84) \quad \text{\texttt{feq:bouhtou_ineq_p_zero}}$$

which is equivalent to Lemma 28.

thm:ssa_ineq

Theorem 29. Let $A, B, C \geq 0$ be positive semidefinite matrices, then

$$\sum_j \lambda_j(A+B) \log \lambda_j(A+B) + \sum_j \lambda_j(B+C) \log \lambda_j(B+C) \quad \text{\texttt{eq:ssa_like_ineq}} \quad (85)$$

$$\leq \sum_j \lambda_j(B) \log \lambda_j(B) + \sum_j \lambda_j(A+B+C) \log \lambda_j(A+B+C). \quad (86)$$

The most comprehensive result concerning the eigenvalues of partial sums of Hermitian matrices we have encountered thus far is [Sag13, Proposition 2.3] which provides the following result.

thm:sagnol_ineq

Theorem 30. Let $A, B, C \geq 0$ be positive semidefinite matrices and let f be a real valued and differentiable function over the non-negative real numbers $\mathbb{R}_{\geq 0}$ such that the derivative of f is an operator monotone function over the positive real numbers $\mathbb{R}_{> 0}$. Then the following inequality holds

eq:sagnol_ineq

$$\text{Tr } f(A+B+C) + \text{Tr } f(B) \geq \text{Tr } f(A+B) + \text{Tr } f(B+C). \quad (87)$$

Proof. The following proof is a rough summary of the proof found in [Sag13, Section 2.1]. Let X, Y be positive semidefinite and consider following difference:

$$F(Y|X) := \text{Tr } f(X+Y) - \text{Tr } f(Y). \quad (88)$$

The inequality in Eq. (87) is thus equivalent to $F(C|B) \leq F(A+C|B)$. To prove Eq. (87) it suffices to prove that the value of $F(Y|X)$ is a non-decreasing function under $Y \mapsto Y+Z$ for arbitrary positive semidefinite X, Y , and Z . To establish the claim, let $\epsilon > 0$ be small and let $Z = \epsilon W$ for non-zero positive semidefinite W . It can be shown⁴ that

$$(\partial_Y F(Y|X))(W) := \lim_{\epsilon \rightarrow 0} \frac{1}{\epsilon} (F(Y + \epsilon W|X) - F(Y|X)) \quad (89)$$

$$= \lim_{\epsilon \rightarrow 0} \frac{1}{\epsilon} (\text{Tr } f(X+Y+\epsilon W) - \text{Tr } f(Y+\epsilon W) - \text{Tr } f(X+Y) + \text{Tr } f(Y)) \quad (90)$$

$$= \text{Tr } [(f'(X+Y) - f'(Y))W]. \quad (91)$$

Using the assumption that f' is an operator monotone together with $X+Y \geq Y$ (because $X \geq 0$), we have $f'(X+Y) \geq f'(Y)$ and thus $f'(X+Y) - f'(Y) \geq 0$. Therefore $(\partial_Y F(Y|X))(W) \geq 0$ for all W and thus $F(Y|X)$ is non-decreasing with respect to $Y \mapsto Y+Z$ as desired. $\square$

As corollaries one obtains some of the previously mentioned inequalities for various functions f satisfying the conditions of the theorem; for example, choosing $f(x) = x \log x$ yields Theorem 29, while choosing $f(x) = -x^p$ for $p \in (0, 1]$ yields Lemma 21, and choosing $f(x) = x^p$ for $p \in (1, 2]$ yields the following.

Corollary 31. Let $A, B, C \geq 0$ be positive semidefinite matrices and let $p \in (1, 2]$. Then

$$\sum_j \lambda_j(A+B)^p + \sum_j \lambda_j(B+C)^p \leq \sum_j \lambda_j(B)^p + \sum_j \lambda_j(A+B+C)^p. \quad (92)$$

TC: On Sept. 18th, I came across an interesting discussion on [mathoverflow](https://mathoverflow.net/questions/257562/a-determinant-inequality) <https://mathoverflow.net/questions/257562/a-determinant-inequality> which appears deeply related to the topic of submodular inequalities. In that post the following result is stated without explicit proof and is based on the theory of so-called “hyperbolic polynomials” p and their real roots λ_i . There are a handful of conjectured inequalities on the page. For example, it seems to prove

$$E_k(B)E_k(A+B+C) \leq E_k(A+B)E_k(B+C) \quad (93)$$

where $E_k(X) = e_k(\text{eig}(X))$ where E_k is the k th elementary symmetric polynomial.

⁴The statement follows, without loss of generality, by assuming that all the matrices involved are positive *definite*. This assumption is permitted because f is assumed continuous and the eigenvalues of a matrix are also continuous functions (see [Sag13] for a detailed explanation).

5.3 The CSW inequality

At this stage, it is worth comparing the inequality in Eq. (85) to a similar inequality derived by Christandl, Şahinoğlu, and Walter [CSW18], which holds for tetrahedral eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ which are non-negative and sufficiently small in the sense that $\lambda_1 + |\lambda| \leq 1$:

$$h(|\gamma|) + |\gamma| S(\underline{\gamma}) + h(|\phi|) + |\phi| S(\underline{\phi}) - h(|\beta|) - |\beta| S(\underline{\beta}) - h(|\epsilon|) - |\epsilon| S(\underline{\epsilon}) \geq 0. \quad \text{\texttt{eq:SSA_like_CSW}} \quad (94)$$

The inequality appearing in Eq. (85) is equivalent to the following by ??:

$$-|\gamma| \log |\gamma| + |\gamma| S(\underline{\gamma}) - |\phi| \log |\phi| + |\phi| S(\underline{\phi}) + |\beta| \log |\beta| - |\beta| S(\underline{\beta}) + |\epsilon| \log |\epsilon| - |\epsilon| S(\underline{\epsilon}) \geq 0. \quad \text{\texttt{eq:SSA_like_ineq_equiv}} \quad (95)$$

The difference between Eq. (94) and Eq. (95) is captured by a formal substitution replacing the binary entropy $h(|\lambda|)$ with the expression $-|\lambda| \log |\lambda|$. We note that it is possible to derive Eq. (95) from Eq. (94) by a down-scaling argument. Specifically, while the set of tetrahedral eigenvalues is closed under scaling all eigenvalues by a non-negative r ,

$$(\alpha, \beta, \gamma, \delta, \epsilon, \phi) \mapsto (r\alpha, r\beta, r\gamma, r\delta, r\epsilon, r\phi), \quad \text{\texttt{eq:tet_scaling}} \quad (96)$$

the form and positivity of Eq. (94) is *not* invariant under scaling by r . Define q_r by the formula

$$q_r := r^{-1} (h(r|\gamma|) + h(r|\phi|) - h(r|\beta|) - h(r|\epsilon|)). \quad (97)$$

Letting $r \in (0, 1)$ be positive and small, we see that

$$r^{-1} h(r|\lambda|) = |\lambda| (1 - \log r) - |\lambda| \log |\lambda| + O(r^2) \quad (98)$$

which implies the following limit holds when $|\gamma| + |\phi| = |\beta| + |\epsilon|$:

$$q_0 := \lim_{r \rightarrow 0} q_r = -|\gamma| \log |\gamma| - |\phi| \log |\phi| + |\beta| \log |\beta| + |\epsilon| \log |\epsilon|. \quad (99)$$

In summary, by scaling down the eigenvalues in Eq. (94) by $r \in (0, 1)$ according to Eq. (96) and then scaling up Eq. (94) by multiplication by r^{-1} and taking the limit as $r \rightarrow \infty$, one recovers Eq. (95). Moreover, it can be shown that $q_r \geq q_0$ for all r and thus Eq. (95) is strictly stronger than Eq. (94). To prove this, we see that the trace equalities imply

$$\begin{aligned} q_r - q_0 &= r^{-1} (1 - r|\beta|) \log(1 - r|\beta|) + r^{-1} (1 - r|\epsilon|) \log(1 - r|\epsilon|) \\ &\quad - r^{-1} (1 - r|\gamma|) \log(1 - r|\gamma|) - r^{-1} (1 - r|\phi|) \log(1 - r|\phi|). \end{aligned} \quad (100)$$

Approximating $-(1-x) \log(1-x)$ by a Taylor series for $x \approx 0$,

$$-(1-x) \log(1-x) = x - \frac{x^2}{2} + O(x^3), \quad (101)$$

produces

$$q_r - q_0 = \frac{r}{2} (|\beta|^2 + |\epsilon|^2 - |\gamma|^2 - |\phi|^2) + O(r^2) = r |\alpha| |\delta| + O(r^2) \quad (102)$$

which implies $q_r > q_0$ and least for sufficiently small r . To prove $q_r > q_0$ holds for all r , by continuity, it therefore suffices to show that the gradient of $q_r - q_0$ with respect to r never changes sign for $r > 0$.

$$\begin{aligned} \partial_r (q_r - q_0) &= \frac{1}{r^2} (\log(1 - r|\gamma|) + \log(1 - r|\phi|) - \log(1 - r|\beta|) - \log(1 - r|\epsilon|)) \\ &= \frac{1}{r^2} \log \left(\frac{(1 - r|\gamma|)(1 - r|\phi|)}{(1 - r|\beta|)(1 - r|\epsilon|)} \right). \end{aligned} \quad (103)$$

Thus the gradient $\partial_r (q_r - q_0)$ vanishes only if the argument of the above logarithm is one, or equivalently when

$$0 = (1 - r|\gamma|)(1 - r|\phi|) - (1 - r|\beta|)(1 - r|\epsilon|) = r^2 (|\gamma| |\phi| - |\beta| |\epsilon|) = r^2 |\alpha| |\delta|. \quad (104)$$

As we are implicitly assuming $|\alpha| \neq 0$ and $|\delta| \neq 0$, we see that the gradient of $q_r - q_0$ never changes when $r > 0$ and thus $q_r - q_0$ is increasing with increasing $r > 0$.

6 Representation theory & eigenvalue estimation

6.1 Schur-Weyl duality & eigenvalue estimation

Our main tool for connecting the eigenvalues of an $n \times n$ Hermitian matrix A to the representation theory of the unitary group $U(n)$ will be to examine the Schur-Weyl decomposition of the k -th tensor power $A^{\otimes k}$. To begin, we recall the Schur-Weyl decomposition of $(\mathbb{C}^n)^{\otimes k}$ as a representation of the group $S_k \times U(n)$.

Proposition 32 (Schur-Weyl duality). *Let k and n be positive integers. Then the decomposition of $(\mathbb{C}^n)^{\otimes k}$ into irreducible subrepresentations of $S_k \times U(n)$, known as the Schur-Weyl decomposition, is given by*

$$(\mathbb{C}^n)^{\otimes k} \cong \bigoplus_{\lambda' \vdash_n k} W_{\lambda'} \otimes V_{\lambda'}^n \quad \text{\texttt{\{eq:schur_weyl_decomp\}}} \quad (105)$$

where the direct summands are indexed by Young diagrams λ' with k boxes and at most n rows, and where $W_{\lambda'}$ supports the irreducible representation of S_k and $V_{\lambda'}^n$ supports the irreducible representation of $U(n)$.

The dimension of the $V_{\lambda'}^n$ is given by Weyl's dimension formula [Hal08; IN66],

$$\dim(V_{\lambda'}^n) = \frac{\Delta_n(\tilde{\lambda}')}{\Delta_n(\rho)}, \quad \text{\texttt{\{eq:weyl_dim\}}} \quad (106)$$

whereas the dimension of $W_{\lambda'}$ may be given by [GW00, Corollary 9.1.5],

$$\dim(W_{\lambda'}) = \frac{|\lambda'|!}{\tilde{\lambda}'_1! \tilde{\lambda}'_2! \cdots \tilde{\lambda}'_n!} \Delta_n(\tilde{\lambda}') \quad \text{\texttt{\{eq:frob_dim\}}} \quad (107)$$

where $\tilde{\lambda}' = \lambda' + \rho$, where $\rho = (n-1, n-2, \dots, 1, 0)$ and where $\Delta_n(x_1, \dots, x_n) = \prod_{1 \leq i < j \leq n} (x_i - x_j)$ is the Vandermonde determinant. Note that $\dim(W_{\lambda'})$ is independent of n as the right hand side of Eq. (107) is independent of n provided n is at least the depth of λ' and coincides with the Hook-length formula [GW00, Corollary 9.1.6].

Definition 33. Given a non-zero positive semidefinite $n \times n$ matrix A , a positive integer x , and a Young diagram $\alpha' \vdash_n x$, let

$$\Gamma[A|\alpha'] := \text{Tr}[\underline{A}^{\otimes a} P_{\alpha'}], \quad (108)$$

where $\underline{A} = \frac{A}{\text{Tr}(A)}$ is the normalization of A and $P_{\alpha'}$ projects onto the isotypic component of $(\mathbb{C}^n)^{\otimes a}$ indexed by α' .

As $P_{\alpha'}$ is $U(n)$ -equivariant, one observes that $\Gamma[A|\alpha']$ depends only on the eigenvalues of A . The following result, known as spectrum or eigenvalue estimation, states that the map $\alpha' \mapsto \Gamma[A|\alpha']$ for A positive semidefinite is concentrated those Young diagrams α' whose shape matches the shape of the eigenvalues $\alpha = \text{eig}(A)$ [Chr08; KW01].

\texttt{\{prop:eig_est\}}

Proposition 34 (Eigenvalue estimation). *Let A be a positive semidefinite matrix with eigenvalues $\alpha = \text{eig}(A)$ and let $\alpha' \vdash_n a$. Then*

$$\Gamma[A|\alpha'] \leq \dim(V_{\alpha'}^n) \exp(-aD(\underline{\alpha}' \parallel \underline{\alpha})), \quad (109)$$

where $D(\underline{\alpha}' \parallel \underline{\alpha})$ denotes the Kullback-Leibler divergence of $\underline{\alpha}'$ from $\underline{\alpha}$.

Proof. See [KW01] for a proof (noting that $\text{Tr}(\underline{A}) = 1$).

□
\texttt{\{prop:HS_measure\}}

Proposition 35. *Let $d\mu(A)$ denote the $U(n)$ -invariant Hilbert Schmidt probability measure over the set of $n \times n$ positive semidefinite trace one matrices and let x be a positive integer. Then*

$$\mathbb{E}_A[A^{\otimes a}] := \int d\mu(A) A^{\otimes k} = \sum_{\alpha' \vdash_n x} q_{\alpha'} P_{\alpha'} \quad \text{\texttt{\{eq:HS_mean\}}} \quad (110)$$

where the coefficient $q_{\alpha'}$ equals

$$q_{\alpha'} = \frac{\dim(V_{\alpha'}^n)}{\dim(W_{\alpha'}) \dim(V_{(a)}^{n^2})} \quad \text{\texttt{\{eq:HS_mean_coeff\}}} \quad (111)$$

Proof. See [ZS01] or [Wal14] for proof details. □

prop:HS_integral_uni

Proposition 36. *Let A be a positive semidefinite $n \times n$ matrix such that $\text{Tr}(A) = 1$. Then*

$$\mathbb{E}_A \Gamma[A|\alpha'] := \int d\mu(A) \Gamma[A|\alpha'] = \frac{\dim(V_{\alpha'}^n)^2}{\dim(V_{(a)}^n)}. \quad (112)$$

prop:completeness_rel_uni

Proposition 37. *Let A be positive semidefinite $n \times n$ matrix with $\text{Tr}(A) = 1$ with eigenvalues $\alpha = \text{eig}(A)$ and rank $r(\alpha) = \text{rank}(A)$. Then for any positive integer a ,*

$$\sum_{\alpha' \vdash_{r(\alpha)} a} \Gamma[A|\alpha'] = 1, \quad (113)$$

where the sum is over Young diagrams α' with size a with depth at most $r(\alpha)$.

6.2 Littlewood-Richardson coefficients & trigonal eigenvalues

The purpose of this section is to reveal how eigenvalue estimation can be used to prove a kind of correspondence between trigonal eigenvalue (α, β, γ) and Littlewood-Richardson coefficients $c_{\alpha'\beta'}^{\gamma'}$. Given a triple of Hermitian matrices (A, B, C) with eigenvalues (α, β, γ) satisfying $A + B = C$, by eigenvalue estimation we see that each of

$$\alpha' \mapsto \Gamma[A|\alpha'], \quad \beta' \mapsto \Gamma[B|\beta'], \quad \gamma' \mapsto \Gamma[C|\gamma'], \quad (114)$$

are respectively concentrated around Young diagrams $(\alpha', \beta', \gamma')$ whose normalized shapes $(\underline{\alpha'}, \underline{\beta'}, \underline{\gamma'})$ approximate the normalized eigenvalues $(\underline{\alpha}, \underline{\beta}, \underline{\gamma})$. As a result of this concentration, one can detect whether there exists Hermitian matrices (A, B, C) with eigenvalues (α, β, γ) satisfying $A + B = C$ by examining if whether there exists an intersection of subspaces onto which $P_{\alpha'}$, $P_{\beta'}$ and $P_{\gamma'}$ project. To support this claim, we define the following quantity.

Definition 38. Given positive semidefinite Hermitian matrices A, B and $C = A + B$ and positive integers x, y where $\alpha \vdash_n x$, $\beta \vdash_n y$, and $\gamma \vdash_n x + y$.

$$\Gamma[A \ B \ C|\alpha' \ \beta' \ \gamma'] := \text{Tr}[(A^{\otimes a} \otimes B^{\otimes b})(P_{\alpha'} \otimes P_{\beta'})P_{\gamma}] \quad (115)$$

6.3 Generalized 6j symbols & tetrahedral eigenvalues

defn:tet_gamma

Definition 39. Given positive integers a, b , and d , let $c = a + b$, $f = b + d$, and $e = a + b + d$. Let $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$ be Young diagrams of the following sizes:

$$\alpha' \vdash a, \quad \beta' \vdash b, \quad \gamma' \vdash c, \quad \delta' \vdash d, \quad \epsilon' \vdash e, \quad \phi' \vdash f. \quad (116)$$

{eq:yd_sizes}

and define corresponding projection operators acting on $(\mathbb{C}^n)^{\otimes a+b+d}$ by

$$\begin{aligned} \Pi_{\alpha'} &:= P_{\alpha'} \otimes \text{id}^{\otimes b} \otimes \text{id}^{\otimes d}, & \Pi_{\beta'} &:= \text{id}^{\otimes a} \otimes P_{\beta'} \otimes \text{id}^{\otimes d}, & \Pi_{\gamma'} &:= P_{\gamma'} \otimes \text{id}^{\otimes d}, \\ \Pi_{\delta'} &:= \text{id}^{\otimes a} \otimes \text{id}^{\otimes b} \otimes P_{\delta'}, & \Pi_{\epsilon'} &:= P_{\epsilon'}, & \Pi_{\phi'} &:= \text{id}^{\otimes a} \otimes P_{\phi'}. \end{aligned} \quad (117)$$

{eq:extended_projs}

Given $n \times n$ positive semidefinite matrices A, B, D with $C = A + B$, $F = B + D$, and $E = A + B + D$, let

$$\Gamma \left[\begin{smallmatrix} A & B & C \\ D & E & F \end{smallmatrix} \middle| \begin{smallmatrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{smallmatrix} \right] = \text{Tr}[(\underline{A}^{\otimes a} \otimes \underline{B}^{\otimes b} \otimes \underline{D}^{\otimes d}) \Pi_{\alpha'} \Pi_{\beta'} \Pi_{\gamma'} \Pi_{\delta'} \Pi_{\epsilon'} \Pi_{\phi'}]. \quad (118)$$

Remark 40. In the analysis that follows, it will be convenient to avoid certain edge cases pertaining to the candidate sextuple of positive semidefinite eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$. Specifically, let the extremal traces be denoted by

$$\tau_{\min} := \min\{|\alpha|, |\beta|, |\gamma|, |\delta|, |\epsilon|, |\phi|\}, \quad \tau_{\max} := \max\{|\alpha|, |\beta|, |\gamma|, |\delta|, |\epsilon|, |\phi|\}, \quad (119)$$

and the extremal ranks be denoted by

$$r_{\min} := \min\{r(\alpha), r(\beta), r(\gamma), r(\delta), r(\epsilon), r(\phi)\}, \quad r_{\max} := \max\{r(\alpha), r(\beta), r(\gamma), r(\delta), r(\epsilon), r(\phi)\}. \quad (120)$$

In order to avoid the edge case where one of the eigenvalues $\lambda \in \{\alpha, \beta, \gamma, \delta, \epsilon, \phi\}$ is totally zero, we sometimes demand that the minimum trace is strictly positive or equivalently that the minimum rank is at least one:

$$\tau_{\min} > 0, \quad r_{\min} \geq 1. \quad (121)$$

Another technical difficulty which complicates subsequent analysis is that the set of projection operators $\{\Pi_{\alpha'}, \Pi_{\beta'}, \Pi_{\gamma'}, \Pi_{\delta'}, \Pi_{\epsilon'}, \Pi_{\phi'}\}$ defined by Eq. (117) is *almost* pairwise mutually commuting. lem:tet_com_rels

Lemma 41. *Every pair of projection operators in $\{\Pi_{\alpha'}, \Pi_{\beta'}, \Pi_{\gamma'}, \Pi_{\delta'}, \Pi_{\epsilon'}, \Pi_{\phi'}\}$ is commuting with only possible exception being $\Pi_{\gamma'}$ and $\Pi_{\phi'}$. Therefore, the operators $\Pi_{\alpha'\beta'\delta'\epsilon'}^{\gamma'}$ and $\Pi_{\alpha'\beta'\delta'\epsilon'}^{\phi'}$ defined by*

$$\Pi_{\alpha'\beta'\delta'\epsilon'}^{\gamma'} = \Pi_{\gamma'}\Pi_{\alpha'}\Pi_{\beta'}\Pi_{\delta'}\Pi_{\epsilon'}, \quad \text{and} \quad \Pi_{\alpha'\beta'\delta'\epsilon'}^{\phi'} = \Pi_{\phi'}\Pi_{\alpha'}\Pi_{\beta'}\Pi_{\delta'}\Pi_{\epsilon'}, \quad (122)$$

are projection operators.

Proof. That all projection operators except the pair $\{\Pi_{\gamma'}, \Pi_{\phi'}\}$ commute follows from elementary applications of Schur's lemma. The obstruction for the pair $\{\Pi_{\gamma'}, \Pi_{\phi'}\}$ is simply that $\Pi_{\gamma'}$ is not necessarily $\text{SU}(n)$ -equivariant with respect to the diagonal action on $\mathbb{C}^{\otimes b+d}$ and likewise $\Pi_{\phi'}$ is not necessarily $\text{SU}(n)$ -equivariant with respect to the diagonal action on $\mathbb{C}^{\otimes a+b}$. $\square$

The following proposition is the analogue of Proposition 36. prop:HS_integral_tet

Proposition 42. *Let $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$ be Young diagrams with sizes given by Eq. (116). Then*

$$\begin{aligned} \mathbb{E}_{A,B,D} \Gamma \left[\begin{array}{ccc|ccc} A & B & C & \alpha' & \beta' & \gamma' \\ D & E & F & \delta' & \epsilon' & \phi' \end{array} \right] &:= \iiint d\mu(A) d\mu(B) d\mu(D) \Gamma \left[\begin{array}{ccc|ccc} A & B & C & \alpha' & \beta' & \gamma' \\ D & E & F & \delta' & \epsilon' & \phi' \end{array} \right] \quad \text{\textcolor{blue}{eq:tet_mean_value}} \\ &= R_{\alpha'\beta'\delta';\epsilon'} \left\| \left\{ \begin{array}{ccc} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{array} \right\} \right\|_{\text{HS}}^2 \quad (123) \end{aligned}$$

where $R_{\alpha'\beta'\delta';\epsilon'}$ is a ratio of dimensions:

$$R_{\alpha'\beta'\delta';\epsilon'} = \frac{\dim(V_{\alpha'}^n) \dim(V_{\beta'}^n) \dim(V_{\delta'}^n) \dim(V_{\epsilon'}^n)}{\dim(V_{(a)}^{n^2}) \dim(V_{(b)}^{n^2}) \dim(V_{(d)}^{n^2})}. \quad (124)$$

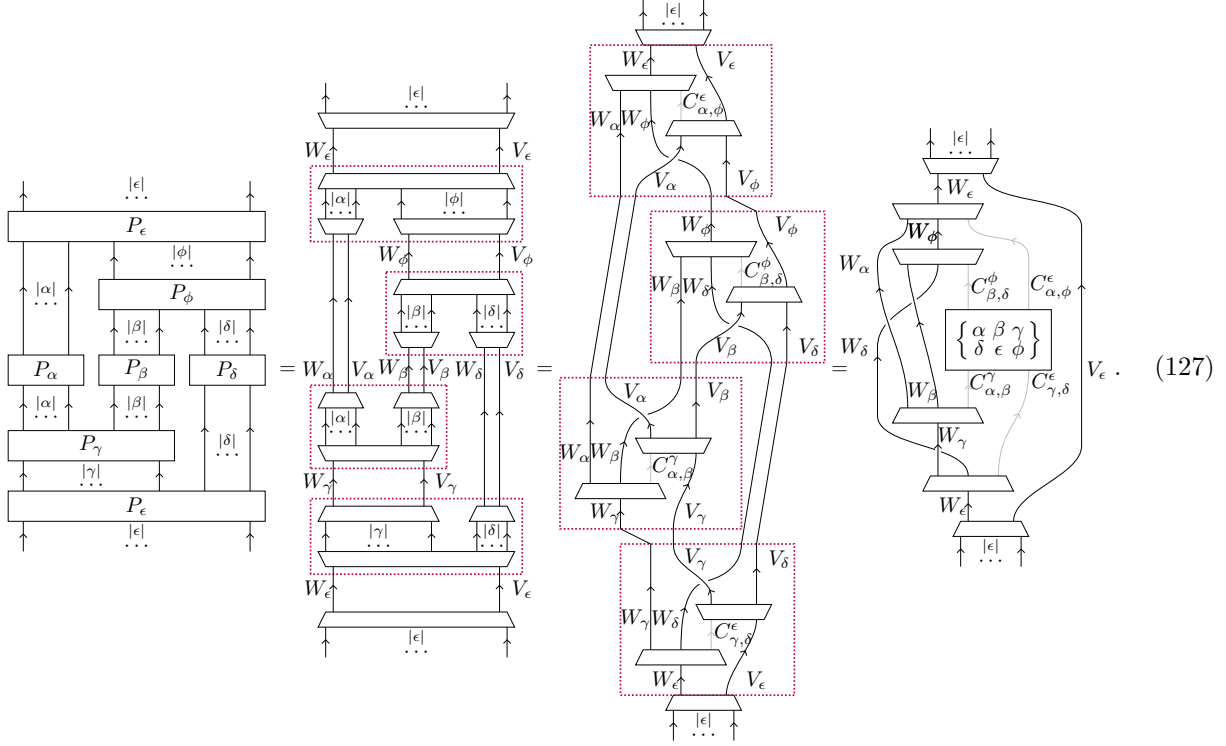
Proof. To prove Eq. (123), we use ?? and Lemma 41 to see that

$$\mathbb{E}_{A,B,D} \Gamma \left[\begin{array}{ccc|ccc} A & B & C & \alpha' & \beta' & \gamma' \\ D & E & F & \delta' & \epsilon' & \phi' \end{array} \right] = q_{\alpha'} q_{\beta'} q_{\delta'} \text{Tr}(\Pi_{\alpha'} \Pi_{\beta'} \Pi_{\gamma'} \Pi_{\delta'} \Pi_{\epsilon'} \Pi_{\phi'}). \quad (125)$$

We claim that the product $\Pi_{\alpha'} \Pi_{\beta'} \Pi_{\gamma'} \Pi_{\delta'} \Pi_{\epsilon'} \Pi_{\phi'}$ is related to the generalized $6j$ operator $\left\{ \begin{array}{ccc} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{array} \right\}$ in that there exists appropriately-typed isometries ι_I and ι_{II} such that

$$\Pi_{\alpha'} \Pi_{\beta'} \Pi_{\gamma'} \Pi_{\delta'} \Pi_{\epsilon'} \Pi_{\phi'} = \iota_{II} \left(\text{id}_{V_{\epsilon'}^n} \otimes \text{id}_{W_{\alpha}} \otimes \text{id}_{W_{\beta}} \otimes \text{id}_{W_{\delta}} \otimes \left\{ \begin{array}{ccc} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{array} \right\} \right) \iota_I^{\dagger} \quad \text{\textcolor{blue}{eq:ud_sixj_prod_proj_iso}} \quad (126)$$

This follows from the follow diagrammatic calculation **TODO: Explain this.:**



From Eq. (126) we conclude both

$$\text{Tr}(\Pi_{\alpha'} \Pi_{\beta'} \Pi_{\gamma'} \Pi_{\delta'} \Pi_{\epsilon'} \Pi_{\phi'}) = \dim(V_\epsilon^n) \dim(W_\alpha) \dim(W_\beta) \dim(W_\delta) \left\| \left\{ \begin{matrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{matrix} \right\} \right\|_{\text{HS}}^{\{\text{eq:nd_six}\} \{\text{prod_proj_hs_norm}\}}, \quad (128)$$

Combining Eq. (128) with Eq. (125) and Eq. (111) then yields Eq. (123). $\square$

The following proposition is the analogue of Proposition 37.

prop:completeness_rel_tet

Proposition 43. Let (A, B, C, D, E, F) be tetrahedral positive semidefinite matrices with eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$. Then

$$\sum_{\substack{\alpha' \beta' \gamma' \\ \delta' \epsilon' \phi'}} \Gamma \left[\begin{matrix} A & B & C \\ D & E & F \end{matrix} \middle| \begin{matrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{matrix} \right] = 1. \quad (129)$$

eq:tet_completeness

where the sum is taken over Young diagrams with the following sizes:

$$\alpha' \vdash_{r(\alpha)} a, \quad \beta' \vdash_{r(\beta)} b, \quad \gamma' \vdash_{r(\gamma)} c, \quad \delta' \vdash_{r(\delta)} d, \quad \epsilon' \vdash_{r(\epsilon)} e, \quad \phi' \vdash_{r(\phi)} f. \quad (130)$$

eq:y_d_sizes_rank

Proof. To begin, note that Eq. (129) holds when the sum is taken over Young diagrams with unrestricted rank according to Eq. (116) because $\sum_{\lambda' \vdash_n \ell} P_{\lambda'} = \text{id}_n^{\otimes \ell}$ for all positive integers ℓ and $\lambda' \in \{\alpha', \beta', \gamma', \delta', \epsilon', \phi'\}$. That one can restrict to Young diagrams with depths bounded according to Eq. (116) follows because $\Gamma \left[\begin{matrix} A & B & C \\ D & E & F \end{matrix} \middle| \begin{matrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{matrix} \right] = 0$ whenever the depth of *any* of $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$ exceeds the corresponding ranks $(r(\alpha), r(\beta), r(\gamma), r(\delta), r(\epsilon), r(\phi))$.⁵ $\square$

Our next goal will be to derive two upper bounds on the absolute value of $\Gamma \left[\begin{matrix} A & B & C \\ D & E & F \end{matrix} \middle| \begin{matrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{matrix} \right]$. Our first upper bound is *independent* of the given matrices and follows as a corollary of Lemma 41.

⁵This claim actually follows from Theorem 45 and will be proven shortly. Importantly, Theorem 45 does not rely on Proposition 43.

Corollary 44. Let $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$ be Young diagrams of sizes given by Eq. (116). Then for all tetrahedral, non-zero positive semidefinite (A, B, C, D, E, F) , we have

$$\left| \Gamma \begin{bmatrix} A & B & C \\ D & E & F \end{bmatrix} \begin{bmatrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{bmatrix} \right| \leq \left\| \left\{ \begin{bmatrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{bmatrix} \right\} \right\|_{\infty}. \quad \text{feq:tet_gamma_size_upper_bound} \quad (131)$$

Proof. From Eq. (122) we have

$$\Gamma \begin{bmatrix} A & B & C \\ D & E & F \end{bmatrix} \begin{bmatrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{bmatrix} = \text{Tr}(\Pi_{\alpha'\beta'\delta'\epsilon'}^{\phi'} (\underline{A}^{\otimes a} \otimes \underline{B}^{\otimes b} \otimes \underline{D}^{\otimes d}) \Pi_{\alpha'\beta'\delta'\epsilon'}^{\gamma'}). \quad (132)$$

Moreover, as $\Omega := \underline{A}^{\otimes a} \otimes \underline{B}^{\otimes b} \otimes \underline{D}^{\otimes d}$ is positive semidefinite trace one, and $\Pi_1 := \Pi_{\alpha'\beta'\delta'\epsilon'}^{\phi'}$ and $\Pi_2 := \Pi_{\alpha'\beta'\delta'\epsilon'}^{\gamma'}$ are projection operators by Lemma 41, we have

$$\left| \Gamma \begin{bmatrix} A & B & C \\ D & E & F \end{bmatrix} \begin{bmatrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{bmatrix} \right| = |\text{Tr}(\Pi_1 \Omega \Pi_2)| \leq |\langle v, \Pi_1 \Pi_2 v \rangle| \leq \|\Pi_1 \Pi_2\|_{\infty} = \left\| \left\{ \begin{bmatrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{bmatrix} \right\} \right\|_{\infty}. \quad (133)$$

where the last equality follows from the decomposition provided by ??.

Our second upper bound on the magnitude of $\Gamma \begin{bmatrix} A & B & C \\ D & E & F \end{bmatrix} \begin{bmatrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{bmatrix}$ depends on the *eigenvalues* $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ of the given tetrahedral matrices (A, B, C, D, E, F) where $C = A + B$, $F = B + D$ and $E = A + B + D$. This result proves that in the limit of large values for (x, y, z) , the magnitude of $\Gamma \begin{bmatrix} A & B & C \\ D & E & F \end{bmatrix} \begin{bmatrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{bmatrix}$ is sharply concentrated around those sextuples of Young diagrams $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$ which resemble the sextuple of eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ of (A, B, C, D, E, F) .

thm:abd_dep_upper_bound

Theorem 45. Let everything be as in Definition 39 and let the Young diagrams $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$ have sizes given by

$$\begin{aligned} \alpha' \vdash a &= [k | \alpha|], & \beta' \vdash b &= [k | \beta|], & \gamma' \vdash c &= [k | \alpha|] + [k | \beta|], \\ \delta' \vdash d &= [k | \delta|], & \epsilon' \vdash e &= [k | \alpha|] + [k | \beta|] + [k | \delta|], & \phi' \vdash f &= [k | \beta|] + [k | \delta|], \end{aligned} \quad \text{feq:ceil_yd_sizes} \quad (134)$$

for some positive integer $k > \tau_{\min}^{-1}$ where $\tau_{\min}$ is the minimum trace. Then

$$\left| \Gamma \begin{bmatrix} A & B & C \\ D & E & F \end{bmatrix} \begin{bmatrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{bmatrix} \right| \leq p(k) \exp(-k I_{\alpha\beta\gamma\delta\epsilon\phi}^{\alpha'\beta'\gamma'\delta'\epsilon'\phi'} / 6). \quad \text{feq:tet_gamma_upper_bound} \quad (135)$$

where $p(k)$ is a polynomial in k and where $I_{\alpha\beta\gamma\delta\epsilon\phi}^{\alpha'\beta'\gamma'\delta'\epsilon'\phi'}$ is a quantity which measures the deviation of the primed quantities from the unprimed quantities:

$$I_{\alpha\beta\gamma\delta\epsilon\phi}^{\alpha'\beta'\gamma'\delta'\epsilon'\phi'} = |\alpha| D(\underline{\alpha}' \| \underline{\alpha}) + |\beta| D(\underline{\beta}' \| \underline{\beta}) + |\gamma| D(\underline{\gamma}' \| \underline{\gamma}) + |\delta| D(\underline{\delta}' \| \underline{\delta}) + |\epsilon| D(\underline{\epsilon}' \| \underline{\epsilon}) + |\phi| D(\underline{\phi}' \| \underline{\phi}). \quad (136)$$

Proof. The first portion of the proof does not depend on the value of k . For now, we suppose the Young diagrams $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$ have sizes determined by Eq. (116) for x, y, z arbitrary positive integers. As in the proof of Corollary 44, we let $\Omega := \underline{A}^{\otimes a} \otimes \underline{B}^{\otimes b} \otimes \underline{D}^{\otimes d}$. Using the commutation relations provided by Lemma 41 we apply the Cauchy-Schwarz inequality to $\langle X, Y \rangle = \text{Tr}(X^* Y) \leq \text{Tr}(X^* X)^{\frac{1}{2}} \text{Tr}(Y^* Y)^{\frac{1}{2}}$ for $X = \Omega^{\frac{1}{2}} \Pi_{\phi'} \Pi_{\alpha'} \Pi_{\beta'}$ and $Y = \Omega^{\frac{1}{2}} \Pi_{\gamma'} \Pi_{\delta'} \Pi_{\epsilon'}$ to obtain

$$\begin{aligned} \left| \Gamma \begin{bmatrix} A & B & C \\ D & E & F \end{bmatrix} \begin{bmatrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{bmatrix} \right|^6 &= |\text{Tr}[(\Pi_{\phi'} \Pi_{\alpha'} \Pi_{\beta'}) \Omega (\Pi_{\gamma'} \Pi_{\delta'} \Pi_{\epsilon'})]|^6 \\ &\leq \text{Tr}(\Pi_{\phi'} \Pi_{\alpha'} \Pi_{\beta'} \Omega)^3 \text{Tr}(\Pi_{\gamma'} \Pi_{\delta'} \Pi_{\epsilon'} \Omega)^3 \\ &\leq \text{Tr}(\Pi_{\phi'} \Omega) \text{Tr}(\Pi_{\alpha'} \Omega) \text{Tr}(\Pi_{\beta'} \Omega) \text{Tr}(\Pi_{\gamma'} \Omega) \text{Tr}(\Pi_{\delta'} \Omega) \text{Tr}(\Pi_{\epsilon'} \Omega), \end{aligned} \quad (137)$$

where the last inequality follows because $\Pi_{\lambda'} \leq \text{id}_n^{|\lambda'|}$ for each λ' in $\{\alpha', \beta', \gamma', \delta', \epsilon', \phi'\}$.

Now, for each $\lambda' \in \{\alpha', \beta', \gamma', \delta', \epsilon', \phi'\}$, it is possible to upper-bound $\text{Tr}(\Pi_{\lambda'} \Omega)$ using the eigenvalue estimation technique (Proposition 34). For a detailed example, we will consider value of $\text{Tr}(\Pi_{\gamma'} \Omega)$:

$$\text{Tr}(\Pi_{\gamma'} \Omega) = \frac{\text{Tr}[(P_{\gamma'} \otimes \text{id}^{\otimes d})(A^{\otimes a} \otimes B^{\otimes b} \otimes D^{\otimes d})]}{|\alpha|^a |\beta|^b |\delta|^d} = \frac{\text{Tr}[P_{\gamma'}(A^{\otimes a} \otimes B^{\otimes b})]}{|\alpha|^a |\beta|^b}. \quad (138)$$

Under the assumption that $C = A + B$, we have $C^{\otimes c} = (A + B)^{\otimes a+b}$ which implies

$$\mathrm{Tr}(P_{\gamma'} C^{\otimes c}) = \sum_{k=0}^c \binom{c}{k} \mathrm{Tr}(P_{\gamma'} (A^{\otimes k} \otimes B^{\otimes c-k})). \quad (139)$$

Here we have used the S_c -equivariance of $P_{\gamma'}$. Since each of the summands above are non-negative, we conclude for all $0 \leq j \leq c$ that

$$\mathrm{Tr}(P_{\gamma'} C^{\otimes c}) \geq \binom{c}{j} \mathrm{Tr}(P_{\gamma'} (A^{\otimes j} \otimes B^{\otimes c-j})). \quad (140)$$

Letting $j = a$ and rearranging yields

$$\mathrm{Tr}(P_{\gamma'} (A^{\otimes a} \otimes B^{\otimes b})) \leq \frac{a!b!}{c!} \mathrm{Tr}(P_{\gamma'} C^{\otimes c}). \quad (141)$$

Combining this result with the spectrum estimation bound (Proposition 34) yields

$$\mathrm{Tr}(\Pi_{\gamma'} \Omega) \leq \frac{|\gamma|^c}{|\alpha|^a |\beta|^b} \frac{a!b!}{c!} \dim(V_{\gamma'}^n) \exp(-cD(\underline{\gamma}' \parallel \underline{\gamma})). \quad (142)$$

Analogous calculations for $\lambda' \in \{\alpha', \beta', \gamma', \delta', \epsilon', \phi'\}$ produce the following set of upper bounds:

$$\begin{aligned} \mathrm{Tr}(\Pi_{\alpha'} \Omega) &\leq \dim(V_{\alpha'}^n) \exp(-aD(\underline{\alpha}' \parallel \underline{\alpha})), \\ \mathrm{Tr}(\Pi_{\beta'} \Omega) &\leq \dim(V_{\beta'}^n) \exp(-bD(\underline{\beta}' \parallel \underline{\beta})), \\ \mathrm{Tr}(\Pi_{\gamma'} \Omega) &\leq \dim(V_{\gamma'}^n) \frac{|\gamma|^c}{|\alpha|^a |\beta|^b} \frac{a!b!}{c!} \exp(-cD(\underline{\gamma}' \parallel \underline{\gamma})), \\ \mathrm{Tr}(\Pi_{\delta'} \Omega) &\leq \dim(V_{\delta'}^n) \exp(-dD(\underline{\delta}' \parallel \underline{\delta})), \\ \mathrm{Tr}(\Pi_{\epsilon'} \Omega) &\leq \dim(V_{\epsilon'}^n) \frac{|\epsilon|^e}{|\alpha|^a |\beta|^b |\delta|^d} \frac{a!b!d!}{e!} \exp(-eD(\underline{\epsilon}' \parallel \underline{\epsilon})), \\ \mathrm{Tr}(\Pi_{\phi'} \Omega) &\leq \dim(V_{\phi'}^n) \frac{|\phi|^f}{|\beta|^b |\delta|^d} \frac{b!d!}{f!} \exp(-fD(\underline{\phi}' \parallel \underline{\phi})). \end{aligned} \quad (143)$$

Our objective now will be to use the k -dependent values for (a, b, c, d, e, f) to upper bound the multinomial prefactors appearing in the bounds for γ' , ϵ' , and ϕ' above in terms of k . From Eq. (134), we obtain the following estimates:

$$\begin{aligned} k|\alpha| &\leq a < k|\alpha| + 1, & k|\beta| &\leq b < k|\beta| + 1, & k|\gamma| &\leq c < k|\gamma| + 2, & \text{\texttt{\textcolor{blue}{\texttt{seq:size_estimates}}}} \\ k|\delta| &\leq d < k|\delta| + 1, & k|\epsilon| &\leq e < k|\epsilon| + 3, & k|\phi| &\leq f < k|\phi| + 2. \end{aligned} \quad (144)$$

An application of Lemma 65 for binomial distributions implies the following bound

$$\frac{|\gamma|^c}{|\alpha|^a |\beta|^b} \frac{a!b!}{c!} \leq \exp(4) (k|\gamma| + 3)^2. \quad (145)$$

Moreover, from Weyl's dimension formula, Eq. (106), we obtain [CM06, Equation 7]:

$$\dim(V_{\gamma'}^n) \leq (|\gamma'| + 1)^{n(n-1)/2} = (c + 1)^{n(n-1)/2}, \quad (146)$$

because $(\gamma'_i - i) - (\gamma'_j - j) \leq |\gamma'| + 1$ as well as $j - i \geq 1$. Altogether, we can relax the bound on $\mathrm{Tr}(\Pi_{\gamma'} \Omega)$ to obtain

$$\mathrm{Tr}(\Pi_{\gamma'} \Omega) \leq \exp(4) (k|\gamma| + 3)^{\frac{n(n-1)}{2} + 2} \exp(-k|\gamma| D(\underline{\gamma}' \parallel \underline{\gamma})). \quad (147)$$

A similar calculations for $\mathrm{Tr}(\Pi_{\epsilon'} \Omega)$ yields (noting to apply Lemma 65 for trinomial distributions) yields:

$$\mathrm{Tr}(\Pi_{\epsilon'} \Omega) \leq \exp(16) (k|\epsilon| + 4)^{\frac{n(n-1)}{2} + 3} \exp(-k|\epsilon| D(\underline{\epsilon}' \parallel \underline{\epsilon})). \quad (148)$$

It is also not too difficult to see that the polynomial in k prefactor in the upper bound on $\text{Tr}(\Pi_\epsilon \Omega)$ above is the largest such prefactor and thus Eq. (135) indeed holds if we take $p(k)$ to be

$$p(k) = \exp(16)(k\tau_{\max} + 4)^{\frac{n(n-1)}{2}+3}. \quad \text{\textcolor{blue}{\{eq:p_of_k\}}} \quad (149)$$

□

While Theorem 45 states that $\Gamma \begin{bmatrix} A & B & C \\ D & E & F \end{bmatrix} \begin{smallmatrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{smallmatrix}$ concentrates in magnitude around particular Young diagrams, it is not directly applicable to the tetrahedral Horn problem simply because its value depends on more than just the eigenvalues of the matrices (A, B, C, D, E, F) . Nevertheless, combining Theorem 45 with the completeness relation (Eq. (129)) yields a necessary condition for $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ to be tetrahedral. \text{\textcolor{blue}{cor:tet_size_lower_bound}}

Corollary 46. *Let $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ be the eigenvalues of non-zero positive semidefinite tetrahedral Hermitian matrices (A, B, C, D, E, F) with minimum trace $\tau_{\min}$. Then there exists a sextuple of sequences of Young diagrams $(\alpha'_k, \beta'_k, \gamma'_k, \delta'_k, \epsilon'_k, \phi'_k)$ indexed by positive integers $k > \tau_{\min}^{-1}$ with sizes given by Eq. (134) satisfying both (i) the convergence condition*

$$I_{\alpha\beta\gamma\delta\epsilon\phi}^{\alpha'_k\beta'_k\gamma'_k\delta'_k\epsilon'_k\phi'_k} \leq \frac{6}{k} \log p(k)q(k) \xrightarrow{k \rightarrow \infty} 0, \quad \text{\textcolor{blue}{\{eq:sextuple_convergence_rate\}}} \quad (150)$$

and (ii) the lower bound

$$\left\| \left\{ \begin{smallmatrix} \alpha'_k & \beta'_k & \gamma'_k \\ \delta'_k & \epsilon'_k & \phi'_k \end{smallmatrix} \right\} \right\|_{\infty} \geq \frac{1}{q(k)}, \quad \text{\textcolor{blue}{\{eq:ud_size_lower_bound\}}} \quad (151)$$

where $p(k)$ is given by Eq. (149) whereas $q(k)$ is given by

$$q(k) = (k\tau_{\max} + 4)^{6(r_{\max}-1)}. \quad \text{\textcolor{blue}{\{eq:q_of_k\}}} \quad (152)$$

Proof. To proof the claim, we construct the sequence implicitly. For each k , it suffices to take $(\alpha'_k, \beta'_k, \gamma'_k, \delta'_k, \epsilon'_k, \phi'_k)$ to be any sextuple of Young diagrams which maximizes the value of $\left| \Gamma \begin{bmatrix} A & B & C \\ D & E & F \end{bmatrix} \begin{smallmatrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{smallmatrix} \right|$ with sizes determined by Eq. (134):

$$(\alpha'_k, \beta'_k, \gamma'_k, \delta'_k, \epsilon'_k, \phi'_k) := \underset{\alpha'\beta'\gamma'\delta'\epsilon'\phi'}{\operatorname{argmax}} \left| \Gamma \begin{bmatrix} A & B & C \\ D & E & F \end{bmatrix} \begin{smallmatrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{smallmatrix} \right|. \quad (153)$$

While this maximal choice may not be unique, it will necessarily satisfy a lower bound due to the completeness relation satisfied by $\Gamma \begin{bmatrix} A & B & C \\ D & E & F \end{bmatrix} \begin{smallmatrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{smallmatrix}$ Eq. (129). Specifically, this maximal choice must satisfy

$$\left| \Gamma \begin{bmatrix} A & B & C \\ D & E & F \end{bmatrix} \begin{smallmatrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{smallmatrix} \right| \geq \frac{1}{q(k)} \quad \text{\textcolor{blue}{\{eq:tet_gamma_lower_bound\}}} \quad (154)$$

where $q(k)$ is any upper bound on the number of possible summands in the completeness relation Eq. (129) viewed as a function of k . As there less than or equal to $(s+1)^{r-1}$ Young diagrams with r rows with size at most s , letting $q(k)$ be given by Eq. (152) ensures Eq. (154) holds because the size (resp. rank) of any of $(\alpha'_k, \beta'_k, \gamma'_k, \delta'_k, \epsilon'_k, \phi'_k)$ is at most $n\tau_{\max} + 3$ (resp. $r_{\max}$). Moreover, applying Eq. (135) yields

$$I_{\alpha\beta\gamma\delta\epsilon\phi}^{\alpha'_k\beta'_k\gamma'_k\delta'_k\epsilon'_k\phi'_k} \leq \frac{6}{k} \log(q(k)p(k)) \quad (155)$$

and thus Eq. (150) holds because $p(k)$ and $q(k)$ are polynomials in k with coefficients depending on the maximum trace $\tau_{\max}$ and maximum rank $r_{\max}$. Also Eq. (151) follows from Eq. (154) combined with the upper bound provided by Eq. (131). □

Remark 47. The convergence condition above (Eq. (150)) in conjunction with the size conditions given by Eq. (134) together ensure that the Young diagrams $(\alpha'_k, \beta'_k, \gamma'_k, \delta'_k, \epsilon'_k, \phi'_k)$ converge to the eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ in the following sense that

$$\begin{aligned} \frac{\alpha'_k}{k} &\rightarrow \alpha, & \frac{\beta'_k}{k} &\rightarrow \beta, & \frac{\gamma'_k}{k} &\rightarrow \gamma, \\ \frac{\delta'_k}{k} &\rightarrow \delta, & \frac{\epsilon'_k}{k} &\rightarrow \epsilon, & \frac{\phi'_k}{k} &\rightarrow \phi, \end{aligned} \quad \text{\textcolor{blue}{\{eq:sextuple_convergence\}}} \quad (156)$$

where $\frac{\lambda'_k}{k} \rightarrow \lambda$ is shorthand for $\lim_{k \rightarrow \infty} (\lambda'_k)_i / k = \lambda_i$ for all $i \in \{1, \dots, n\}$.

While the existence of a convergent sequence $(\alpha'_k, \beta'_k, \gamma'_k, \delta'_k, \epsilon'_k, \phi'_k)$ of primed Young diagrams is necessary for $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ to be tetrahedral by Corollary 46, our next result proves it is also sufficient. In other words, if $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ are not tetrahedral, then no such sequence can exist.

Corollary 48. *Suppose $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ is a non-tetrahedral sextuple of positive semidefinite and non-zero eigenvalues with minimum trace $\tau_{\min}$. Then for all sequences of Young diagrams $(\alpha'_k, \beta'_k, \gamma'_k, \delta'_k, \epsilon'_k, \phi'_k)$ indexed by $k > \tau_{\min}^{-1}$ with sizes given by Eq. (134) and converging to $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ in the sense that Eq. (150) holds, we have*

$$\left\| \left\{ \begin{smallmatrix} \alpha'_k & \beta'_k & \gamma'_k \\ \delta'_k & \epsilon'_k & \phi'_k \end{smallmatrix} \right\} \right\|_{\text{HS}}^2 \leq \exp \left(-\frac{k\tau_{\min}}{12} D_*^{\alpha\beta\gamma\delta\epsilon\phi} + O(\log k) \right), \quad \text{\textcolor{blue}{eq:seq_ud_sizj_upper_bound}} \quad (157)$$

where $D_*^{\alpha\beta\gamma\delta\epsilon\phi} > 0$ is a positive constant which measures the distance from $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ to the closed set of tetrahedral eigenvalues $(\tilde{\alpha}, \tilde{\beta}, \tilde{\gamma}, \tilde{\delta}, \tilde{\epsilon}, \tilde{\phi})$ with matching traces (i.e., $|\alpha| = |\tilde{\alpha}|$, $|\beta| = |\tilde{\beta}|$, etc.):

$$D_*^{\alpha\beta\gamma\delta\epsilon\phi} = \min_{\substack{\tilde{\alpha}\tilde{\beta}\tilde{\gamma}\tilde{\delta}\tilde{\epsilon}\tilde{\phi} \\ \text{tetrahedral} \\ \forall \lambda: |\tilde{\lambda}| = |\lambda|}} \left(\|\underline{\alpha} - \tilde{\alpha}\|_1^2 + \|\underline{\beta} - \tilde{\beta}\|_1^2 + \|\underline{\gamma} - \tilde{\gamma}\|_1^2 + \|\underline{\delta} - \tilde{\delta}\|_1^2 + \|\underline{\epsilon} - \tilde{\epsilon}\|_1^2 + \|\underline{\phi} - \tilde{\phi}\|_1^2 \right). \quad (158)$$

Proof. Using Eq. (123), we have

$$R_{\alpha'\beta'\delta';\epsilon'} \left\| \left\{ \begin{smallmatrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{smallmatrix} \right\} \right\|_{\text{HS}}^2 = \iiint d\mu(A) d\mu(B) d\mu(D) \Gamma \left[\begin{smallmatrix} A & B & C \\ D & E & F \end{smallmatrix} \middle| \begin{smallmatrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{smallmatrix} \right] \quad (159)$$

where the integral is taken over a triple of $n \times n$ trace one positive semidefinite matrices A, B, D . For each integrand, define a tetrahedral sextuple $(\tilde{A}, \tilde{B}, \tilde{C}, \tilde{D}, \tilde{E}, \tilde{F})$ as follows

$$\begin{aligned} \tilde{A} &:= |\alpha| A, & \tilde{B} &:= |\beta| B, & \tilde{C} &:= |\alpha| A + |\beta| B, \\ \tilde{D} &:= |\delta| D, & \tilde{E} &:= |\alpha| A + |\beta| B + |\delta| D, & \tilde{F} &:= |\beta| B + |\delta| D. \end{aligned} \quad \text{\textcolor{blue}{eq:tilde_defs}} \quad (160)$$

Furthermore, let $(\tilde{\alpha}, \tilde{\beta}, \tilde{\gamma}, \tilde{\delta}, \tilde{\epsilon}, \tilde{\phi})$ be the eigenvalues of $(\tilde{A}, \tilde{B}, \tilde{C}, \tilde{D}, \tilde{E}, \tilde{F})$. In this notation, the integrand above is also equal to $\Gamma \left[\begin{smallmatrix} \tilde{A} & \tilde{B} & \tilde{C} \\ \tilde{D} & \tilde{E} & \tilde{F} \end{smallmatrix} \middle| \begin{smallmatrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{smallmatrix} \right]$. Moreover, by Eq. (135) we have a bound on the integrand,

$$\left| \Gamma \left[\begin{smallmatrix} \tilde{A} & \tilde{B} & \tilde{C} \\ \tilde{D} & \tilde{E} & \tilde{F} \end{smallmatrix} \middle| \begin{smallmatrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{smallmatrix} \right] \right| \leq p(k) \exp(-k I_{\tilde{\alpha}\tilde{\beta}\tilde{\gamma}\tilde{\delta}\tilde{\epsilon}\tilde{\phi}}^{\alpha'\beta'\gamma'\delta'\epsilon'\phi'} / 6). \quad (161)$$

Now by construction, the eigenvalues $(\tilde{\alpha}, \tilde{\beta}, \tilde{\gamma}, \tilde{\delta}, \tilde{\epsilon}, \tilde{\phi})$ are tetrahedral while by assumption, the eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ are not tetrahedral (although they have matching traces due to Eq. (160)). Therefore we conclude that

$$(\alpha, \beta, \gamma, \delta, \epsilon, \phi) \neq (\tilde{\alpha}, \tilde{\beta}, \tilde{\gamma}, \tilde{\delta}, \tilde{\epsilon}, \tilde{\phi}) \quad (162)$$

and therefore $I_{\tilde{\alpha}\tilde{\beta}\tilde{\gamma}\tilde{\delta}\tilde{\epsilon}\tilde{\phi}}^{\alpha\beta\gamma\delta\epsilon\phi} > 0$.

Our goal now will be to show that for any sequence of Young diagrams $(\alpha'_k, \beta'_k, \gamma'_k, \delta'_k, \epsilon'_k, \phi'_k)$ converging to the non-tetrahedral eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$, one can also guarantee that for sufficiently large k , the exponent $I_{\tilde{\alpha}\tilde{\beta}\tilde{\gamma}\tilde{\delta}\tilde{\epsilon}\tilde{\phi}}^{\alpha'_k\beta'_k\gamma'_k\delta'_k\epsilon'_k\phi'_k}$ above is also bounded from below by a positive constant. Let λ be a dummy variable indexing $\lambda \in \{\alpha, \beta, \gamma, \delta, \epsilon, \phi\}$. By Pinsker's inequality for the relative entropy, we obtain a sequence of lower bounds on $I_{\alpha\beta\gamma\delta\epsilon\phi}^{\alpha'\beta'\gamma'\delta'\epsilon'\phi'}$:

$$I_{\alpha\beta\gamma\delta\epsilon\phi}^{\alpha'\beta'\gamma'\delta'\epsilon'\phi'} = \sum_{\lambda} |\lambda| D(\lambda' \| \lambda) \geq \frac{1}{2} \sum_{\lambda} |\lambda| \|\lambda' - \lambda\|_1^2 \geq \frac{\tau_{\min}}{2} \sum_{\lambda} \|\lambda' - \lambda\|^2 \geq \frac{\tau_{\min}}{2} \max_{\lambda} \|\lambda' - \lambda\|_1^2, \quad (163)$$

where $\tau_{\min}$ is the minimum trace over the eigenvalues (the lower indices). Applying the last bound to $I_{\alpha\beta\gamma\delta\epsilon\phi}^{\alpha'_k\beta'_k\gamma'_k\delta'_k\epsilon'_k\phi'_k}$ yields

$$\max_{\lambda} \|\lambda'_k - \lambda\|_1 \leq \sqrt{2\tau_{\min}^{-1} I_{\alpha\beta\gamma\delta\epsilon\phi}^{\alpha'_k\beta'_k\gamma'_k\delta'_k\epsilon'_k\phi'_k}} \quad (164)$$

whereas applying the second to last bound to $I_{\tilde{\alpha}\tilde{\beta}\tilde{\gamma}\tilde{\delta}\tilde{\epsilon}\tilde{\phi}}^{\alpha'_k\beta'_k\gamma'_k\delta'_k\epsilon'_k\phi'_k}$ (noting that $\tau_{\min}$ is also the minimum trace of $(\tilde{\alpha}, \tilde{\beta}, \tilde{\gamma}, \tilde{\delta}, \tilde{\epsilon}, \tilde{\phi})$) yields

$$\sum_{\lambda} \left\| \underline{\lambda}'_k - \tilde{\lambda} \right\|_1^2 \leq 2\tau_{\min}^{-1} I_{\tilde{\alpha}\tilde{\beta}\tilde{\gamma}\tilde{\delta}\tilde{\epsilon}\tilde{\phi}}^{\alpha'_k\beta'_k\gamma'_k\delta'_k\epsilon'_k\phi'_k}. \quad (165)$$

Furthermore, by the reverse triangle inequality we have

$$\sum_{\lambda} \left\| \underline{\lambda}'_k - \tilde{\lambda} \right\|_1^2 \geq \sum_{\lambda} \left(\left\| \underline{\lambda}'_k - \underline{\lambda} \right\|_1 - \left\| \underline{\lambda} - \tilde{\lambda} \right\|_1 \right)^2 \geq \sum_{\lambda} \left\| \underline{\lambda} - \tilde{\lambda} \right\|_1^2 - 2 \left\| \underline{\lambda}'_k - \underline{\lambda} \right\|_1 \left\| \underline{\lambda} - \tilde{\lambda} \right\|_1. \quad (166)$$

Also by the definition of $D_*^{\alpha\beta\gamma\delta\epsilon\phi}$ we have

$$D_*^{\alpha\beta\gamma\delta\epsilon\phi} \leq \sum_{\lambda} \left\| \underline{\lambda} - \tilde{\lambda} \right\|_1^2. \quad (167)$$

Since $\underline{\lambda}$ and $\tilde{\lambda}$ are both probability distributions we also have $\left\| \underline{\lambda} - \tilde{\lambda} \right\|_1 \leq 2$. Also, $6 \max_{\lambda} \left\| \underline{\lambda}'_k - \underline{\lambda} \right\|_1 \geq \sum_{\lambda} \left\| \underline{\lambda}'_k - \underline{\lambda} \right\|_1$ as there are six summands. Combining all these bounds together yields

$$\left\| \left\{ \begin{smallmatrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{smallmatrix} \right\} \right\|_{\text{HS}}^2 \leq \exp \left(-k \left\{ \frac{\tau_{\min} D_*^{\alpha\beta\gamma\delta\epsilon\phi}}{12} - 2\sqrt{2\tau_{\min} I_{\alpha\beta\gamma\delta\epsilon\phi}^{\alpha'_k\beta'_k\gamma'_k\delta'_k\epsilon'_k\phi'_k}} - \frac{1}{k} \log(R_{\alpha'_k, \beta'_k, \delta'_k, \epsilon'_k}^{-1} p(k)) \right\} \right). \quad (168)$$

By assumption, the Young diagrams $(\alpha'_k, \beta'_k, \gamma'_k, \delta'_k, \epsilon'_k, \phi'_k)$ converge in normalization to the eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ in the sense that $I_{\alpha\beta\gamma\delta\epsilon\phi}^{\alpha'_k\beta'_k\gamma'_k\delta'_k\epsilon'_k\phi'_k}$ tends to zero as $k \rightarrow \infty$ at a rate given by Eq. (150). Moreover, $R_{\alpha'_k, \beta'_k, \delta'_k, \epsilon'_k}^{-1}$ can be upper bounded by a polynomial in k and $p(k)$ is a polynomial in k so Eq. (157) holds as claimed. $\square$

6.4 The large n limit & an entropic inequality

The purpose of this section is to show how to use the necessary condition for $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ to be tetrahedral provided by Corollary 46 to derive inequalities satisfied by tetrahedral eigenvalues. Our strategy will be to prove an upper bound the generalized $6j$ symbol which depends only on the dimensions of the underlying irreducible representations and the multiplicity spaces between them. By combining this upper bound with the lower bound provided by Corollary 46, we obtain a new constraint on the convergent sequence of Young diagrams $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$.

thm:sixj_ssa_bound

Theorem 49. *Let $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$ be a sextuple of Young diagrams. Then*

$$\left\| \left\{ \begin{smallmatrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{smallmatrix} \right\} \right\|_{\text{HS}}^2 \leq \left(c_{\alpha', \phi'}^{\epsilon'} c_{\beta', \delta'}^{\phi'} c_{\alpha', \beta'}^{\gamma'} c_{\gamma', \delta'}^{\epsilon'} \right)^{\frac{1}{2}} \frac{\dim(V_{\gamma'}^n) \dim(V_{\phi'}^n)}{\dim(V_{\beta'}^n) \dim(V_{\epsilon'}^n)}. \quad (169)$$

Proof. Applying both the Cauchy-Schwarz inequality with respect to the contraction annotated below, and

Schur's lemma multiple times yields the following inequality:

$$\begin{aligned}
& \left(\epsilon \right)^2 C_{\alpha, \beta}^{\gamma} C_{\alpha, \phi}^{\epsilon} C_{\beta, \delta}^{\phi} C_{\gamma, \delta}^{\epsilon} = \\
& \left[\begin{array}{c} \text{Young diagram 1} \\ \text{Young diagram 2} \end{array} \right] = \left[\begin{array}{c} \text{Young diagram 3} \\ \text{Young diagram 4} \end{array} \right] \\
& \leq \left[\begin{array}{c} \text{Young diagram 5} \\ \text{Young diagram 6} \end{array} \right]^{\frac{1}{2}} \left[\begin{array}{c} \text{Young diagram 7} \\ \text{Young diagram 8} \end{array} \right]^{\frac{1}{2}} \\
& = \frac{1}{(\beta \epsilon)} \left[\begin{array}{c} \text{Young diagram 9} \\ \text{Young diagram 10} \end{array} \right]^{\frac{1}{2}} \\
& = \frac{\left(\epsilon \right)^2 \left(\gamma \right) \left(\phi \right)}{\left(\beta \right) \left(\epsilon \right)} \left[\begin{array}{c} \text{Young diagram 11} \\ \text{Young diagram 12} \\ \text{Young diagram 13} \\ \text{Young diagram 14} \end{array} \right]^{\frac{1}{2}}
\end{aligned} \tag{170}$$

TODO: Explain what this all means carefully. □

An immediate application of Theorem 49 is to combine it with Corollary 46. If $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ are tetrahedral eigenvalues, then by Corollary 46 there exists a converging sequence of Young diagrams $(\alpha'_k, \beta'_k, \gamma'_k, \delta'_k, \epsilon'_k, \phi'_k)$ such that⁶

$$\frac{1}{q(k)^2} \leq \left\| \left\{ \begin{array}{c} \alpha'_k \beta'_k \gamma'_k \\ \delta'_k \epsilon'_k \phi'_k \end{array} \right\} \right\|_{\text{HS}}^2 \leq \left(c_{\alpha'_k, \phi'_k}^{\epsilon'_k} c_{\beta'_k, \delta'_k}^{\phi'_k} c_{\alpha'_k, \beta'_k}^{\gamma'_k} c_{\gamma'_k, \delta'_k}^{\epsilon'_k} \right)^{\frac{1}{2}} \frac{\dim(V_{\gamma'_k}^n) \dim(V_{\phi'_k}^n)}{\dim(V_{\beta'_k}^n) \dim(V_{\epsilon'_k}^n)}. \tag{171}$$

Remarkably, both $q(k)$ (given by Eq. (152)) and the Littlewood-Richardson coefficients $c_{\alpha'_k, \phi'_k}^{\epsilon'_k}, c_{\beta'_k, \delta'_k}^{\phi'_k}, c_{\alpha'_k, \beta'_k}^{\gamma'_k}$, and $c_{\gamma'_k, \delta'_k}^{\epsilon'_k}$ are *independent* of the dimension parameter n defining the group $U(n)$ (assuming n is at least as large as the maximum depth of any of the involved Young diagrams, i.e., $n \geq r_{\max}$ where $r_{\max}$ is the maximum rank of any of the tetrahedral eigenvalues). This observation provides one with an opportunity to optimize the above bound by varying the value of n .

In particular, it will be useful to better understand how the dimension, $\dim(V_{\lambda'}^n)$, of the irreducible representation $V_{\lambda'}^n$ of $U(n)$ grows as n becomes much larger than the number of rows of λ' . In general, for fixed λ' , the function $n \mapsto \dim(V_{\lambda'}^n)$ can grow at an exponential rate with increasing n . One can find estimates for $\dim(V_{\lambda'}^n)$ in the regime where n is large compared to $|\lambda'|$ and also the last row $\lambda'_{\min}$ is large compared to the depth of λ' in paper by Itzykson and Nauenberg [IN66, Section III]. Here we use an elegant presentation of this scaling, namely [Har23, Lemma 2], which relates the asymptotics of $\dim(V_{\lambda'}^n)$ to $\dim(W_{\lambda'})$.

⁶Here we used the well-known inequality $\|X\|_{\infty} \leq \|X\|_{\text{HS}}$.

prop:harrow_formula

Proposition 50. *Let λ' be a Young diagram with depth at most n . Then*

$$\frac{\dim(V_{\lambda'}^n)}{n^{|\lambda'|}} = \frac{\dim(W_{\lambda'})}{|\lambda'|!} \prod_{(i,j) \in \lambda'} \left(1 + \frac{j-i}{n}\right). \quad (172)$$

Proof. The statement follows from the dimension formulas for $\dim(V_{\lambda'}^n)$ and $\dim(W_{\lambda'})$ in Eq. (106) and Eq. (107). See also [Har23, Appendix A]. $\square$

From Proposition 50 we obtain the following corollary.

cor:harrow_ssa_lim

Corollary 51. *Let $\gamma', \phi', \beta', \epsilon'$ be a quadruple of Young diagrams such that*

$$|\gamma'| + |\phi'| = |\beta'| + |\epsilon'|. \quad (173)$$

{eq:balanced_sizes}

Then

$$\lim_{n \rightarrow \infty} \frac{\dim(V_{\gamma'}^n) \dim(V_{\phi'}^n)}{\dim(V_{\beta'}^n) \dim(V_{\epsilon'}^n)} = \frac{|\beta'|! |\epsilon'|! \dim(W_{\gamma'}) \dim(W_{\phi'})}{|\gamma'|! |\phi'|! \dim(W_{\beta'}) \dim(W_{\epsilon'})} \quad (174)$$

Proof. From Proposition 50 one evidently has for any Young diagram λ' the limit

$$\lim_{n \rightarrow \infty} \frac{\dim(V_{\lambda'}^n)}{n^{|\lambda'|}} = \frac{\dim(W_{\lambda'})}{|\lambda'|!}. \quad (175)$$

The claim then follows because the missing prefactor, $\frac{n^{|\beta'|} n^{|\epsilon'|}}{n^{|\gamma'|} n^{|\phi'|}}$, is equal to one by Eq. (173). $\square$

We are now in a position to prove the main result of this section.

Theorem 52. *Let $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ be tetrahedral eigenvalues. Then*

$$I(\gamma, \phi; \beta, \epsilon) = s(\gamma) + s(\phi) - s(\beta) - s(\epsilon) \geq 0, \quad (176)$$

where

$$s(\lambda) := - \sum_{i=1}^n \lambda_i \log \lambda_i. \quad (177)$$

{eq:little_s_defn}

Proof. As $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ are tetrahedral, the trace conditions (Eq. (9)) necessarily imply that Eq. (173) holds. Therefore, by Corollary 51 and the discussion surrounding the bound in Eq. (171) we conclude that along the convergent sequence of Young diagrams satisfies

$$I' := \lim_{k \rightarrow \infty} \frac{1}{k} \log \left[\frac{|\beta'_k|! |\epsilon'_k|! \dim(W_{\gamma'_k}) \dim(W_{\phi'_k})}{|\gamma'_k|! |\phi'_k|! \dim(W_{\beta'_k}) \dim(W_{\epsilon'_k})} \right] \geq 0 \quad (178)$$

provided the limit I' exists. An application of Stirling's approximation [Rob55] to the dimension formula for $\dim(W_{\lambda'_k})$ together with the approximation $\lambda'_k \approx k\lambda$ yields

$$\frac{1}{k} \frac{\dim(W_{\lambda'_k})}{|\lambda'|!} = s(\lambda) + |\lambda| (1 - \log k) + O(k^{-1} \log k) \quad (179)$$

where $s(\lambda)$ is given in Eq. (177). Moreover by Eq. (173), the $(1 - \log k)$ terms cancel and therefore $I' = I(\gamma, \phi; \beta, \epsilon) \geq 0$ as claimed above. $\square$

7 OLD CONTENT

7.1 Littlewood-Richardson coefficients & 6j symbols

Definition 53 (Littlewood-Richardson coefficients). Let $\alpha, \beta, \gamma \in \mathbb{Z}_{\text{dec}}^n$ be a triple of dominant weights for $\text{GL}(n)$ and let $V_\alpha, V_\beta, V_\gamma$ be the triple of irreducible representations of $\text{GL}(n)$ with corresponding highest weights. Let $C_{\alpha, \beta}^\gamma$ denote the multiplicity space of $\text{GL}(n)$ -intertwiners from V_γ to $V_\alpha \otimes V_\beta$:

$$C_{\alpha, \beta}^\gamma := \text{Hom}_{\text{GL}(n)}(V_\gamma, V_\alpha \otimes V_\beta) \simeq \text{Inv}_{\text{GL}(n)}(V_{\gamma^*} \otimes V_\alpha \otimes V_\beta). \quad (180)$$

The *Littlewood-Richardson coefficient* is defined as the dimension of this multiplicity space:

$$c_{\alpha, \beta}^\gamma := \dim(C_{\alpha, \beta}^\gamma). \quad (181)$$

Definition 54 (Clebsch-Gordon isometries). Given α, β, γ as in ??, a *Clebsch-Gordon isometry* is a $\text{GL}(n)$ -equivariant isometric linear map sending $f \otimes v \in C_{\alpha, \beta}^\gamma \otimes V_\gamma$ to $f(v) \in V_\alpha \otimes V_\beta$ ⁷. Let

$$Y_{\alpha, \beta}^\gamma : C_{\alpha, \beta}^\gamma \otimes V_\gamma \rightarrow V_\alpha \otimes V_\beta, \quad (182)$$

be a Clebsch-Gordon isometry and depict it diagrammatically by

$$\begin{array}{c} V_\alpha \\ \uparrow \\ \text{---} \circ \text{---} \\ \uparrow \quad \uparrow \\ C_{\alpha, \beta}^\gamma \quad V_\gamma \end{array} = \begin{array}{c} V_\alpha \quad V_\beta \\ \uparrow \quad \uparrow \\ \boxed{Y_{\alpha, \beta}^\gamma} \\ \uparrow \quad \uparrow \\ C_{\alpha, \beta}^\gamma \quad V_\gamma \end{array}. \quad (183)$$

The conjugate transpose of the Clebsch-Gordon isometry is

$$Y_{\alpha, \beta}^{\gamma\dagger} : V_\alpha \otimes V_\beta \rightarrow C_{\alpha, \beta}^\gamma \otimes V_\gamma, \quad (184)$$

will be depicted diagrammatically by

$$\begin{array}{c} C_{\alpha, \beta}^\gamma \quad V_\gamma \\ \uparrow \quad \uparrow \\ \text{---} \circ \text{---} \\ \uparrow \quad \uparrow \\ V_\alpha \quad V_\beta \end{array} = \begin{array}{c} C_{\alpha, \beta}^\gamma \quad V_\gamma \\ \uparrow \quad \uparrow \\ \boxed{Y_{\alpha, \beta}^{\gamma\dagger}} \\ \uparrow \quad \uparrow \\ V_\alpha \quad V_\beta \end{array}. \quad (185)$$

Remark 55. In this section, we reviewed the correspondence between (i) the existence of $n \times n$ positive semidefinite matrices (A, B, C) with eigenvalues (α, β, γ) satisfying $A + B = C$ and (ii) the positivity of corresponding Littlewood-Richardson coefficients $c_{\alpha, \beta}^\gamma$ which govern the branching problem for the diagonal subgroup $\text{GL}(n) \hookrightarrow \text{GL}(n) \times \text{GL}(n)$ sending X to (X, X) . Recalling Theorem 17, there is a further correspondence to (iii) the existence of a $2n \times 2n$ positive semidefinite matrix Σ with eigenvalues γ and $n \times n$ blocks A', B' with eigenvalues α, β . One may wonder how to show the correspondence between (ii) and (iii) *directly* without passing through (i). This can be accomplished by considering the branching multiplicities $m_\gamma^{\alpha\beta}$ for the blockwise subgroup $\text{GL}(a) \times \text{GL}(b) \hookrightarrow \text{GL}(a+b)$ sending (A, B) to $A \oplus B$. In fact there exists a representation-theoretic analogue of Theorem 17, specifically [GW00, Theorem 9.2.3]. In particular, the values of $m_\gamma^{\alpha\beta}$ and $c_{\alpha, \beta}^\gamma$ coincide provided the Young diagrams (α, β, γ) involved have depths at most $(\min\{a, n\}, \min\{b, n\}, \min\{a+b, n\})$; this ensures that $V_\alpha^a, V_\beta^b, V_\gamma^{a+b}$ and $V_\alpha^n, V_\beta^n, V_\gamma^n$ are all non-zero representation spaces and therefore $m_\gamma^{\alpha\beta}$ and $c_{\alpha, \beta}^\gamma$ are both meaningfully defined. The distinction between $m_\gamma^{\alpha\beta}$ and $c_{\alpha, \beta}^\gamma$ lies with the fact that it is possible for $c_{\alpha, \beta}^\gamma$ to be zero while $m_\gamma^{\alpha\beta}$ is non-zero (this can happen when the depth of γ is greater than n but at most $a+b$ so that V_γ^n is zero but V_γ^{a+b} is non-zero). This parallels the fact that, in general, the rank of the $2n \times 2n$ matrix Σ in Theorem 17 could be greater than n while the rank of the $n \times n$ matrix C is at most n .

⁷If $C_{\alpha, \beta}^\gamma = 0$, then $Y_{\alpha, \beta}^\gamma$ is identified with the origin in $V_\alpha \otimes V_\beta$.

Definition 56 (Recoupling operators). Let $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$ be Young diagrams of sizes given by Eq. (116). Then we define the $U(n)$ -6j operator to be the operator

$$\left\{ \begin{array}{ccc} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{array} \right\} : C_{\alpha', \beta'}^{\gamma'} \otimes C_{\delta', \gamma'}^{\epsilon'} \rightarrow C_{\alpha', \phi'}^{\epsilon'} \otimes C_{\beta', \delta'}^{\phi'}, \quad (186)$$

defined by

$$\left\{ \begin{array}{ccc} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{array} \right\} := \frac{\text{Tr}_{V_{\epsilon'}} [Y_{\alpha', \phi'}^{\epsilon' \dagger} (\text{id}_{V_{\alpha'}} \otimes Y_{\beta', \delta'}^{\phi' \dagger}) (Y_{\alpha', \beta'}^{\gamma'} \otimes \text{id}_{V_{\delta'}}) Y_{\gamma', \delta'}^{\epsilon'}]}{\dim(V_{\epsilon'})}, \quad (187)$$

which is depicted diagrammatically by

$$\left\{ \begin{array}{ccc} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{array} \right\} := \frac{1}{V_{\epsilon}} \left(\begin{array}{c} C_{\alpha, \phi}^{\epsilon} \\ \uparrow \\ Y_{\alpha, \phi}^{\epsilon \dagger} \\ \downarrow V_{\phi} \\ Y_{\beta, \delta}^{\phi \dagger} \\ \downarrow V_{\beta} \\ Y_{\alpha, \beta}^{\gamma} \\ \downarrow V_{\gamma} \\ Y_{\gamma, \delta}^{\epsilon} \\ \downarrow V_{\delta} \\ C_{\gamma, \delta}^{\epsilon} \end{array} \right) = \frac{1}{V_{\epsilon}} \left(\begin{array}{c} C_{\alpha, \phi}^{\epsilon} \\ \uparrow \\ V_{\alpha} \\ \downarrow \\ C_{\alpha, \beta}^{\gamma} \end{array} \right) \left(\begin{array}{c} C_{\beta, \delta}^{\phi} \\ \uparrow \\ V_{\beta} \\ \downarrow \\ C_{\gamma, \delta}^{\epsilon} \end{array} \right) \left(\begin{array}{c} C_{\alpha, \phi}^{\epsilon} \\ \uparrow \\ V_{\alpha} \\ \downarrow \\ C_{\alpha, \beta}^{\gamma} \end{array} \right) \left(\begin{array}{c} C_{\beta, \delta}^{\phi} \\ \uparrow \\ V_{\beta} \\ \downarrow \\ C_{\gamma, \delta}^{\epsilon} \end{array} \right). \quad (188)$$

WARNING: The convention I am using for the curly braces seems to differ from many of the uses of the curly braces to define the Wigner 6j coefficient which differs in that I think the Wigner 6j coefficient does not include the additional factor of $\dim(V_{\epsilon})$. This lack of common convention seems to be echoed by Coquereaux [Coq23].

Subsequent sections will primarily be interested in the square of the Hilbert-Schmidt norm of the recoupling operator in ???. While the individual coefficients of the recoupling operator are dependent on a choice of basis for each of the multiplicity spaces, its Hilbert-Schmidt and operator norms are not dependent on such a choice of basis. The following lemma provides a formula for the square of the Hilbert-Schmidt norm as an integral over $SU(n)^{\times 4}$.

Lemma 57 (Character formula). Let $\left\{ \begin{array}{ccc} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{array} \right\} : C_{\alpha, \beta}^{\mu} \otimes C_{\mu, \gamma}^{\lambda} \rightarrow C_{\alpha, \nu}^{\lambda} \otimes C_{\beta, \gamma}^{\nu}$ be the recoupling operator defined in ???. Then

$$R_{\delta \epsilon \phi}^{\alpha \beta \gamma} := \dim(V_{\epsilon})^2 \left\| \left\{ \begin{array}{ccc} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{array} \right\} \right\|_{\text{HS}}^2 = \int_{SU(n)^4} du_1 du_2 du_3 du_4 X_{\delta, \epsilon, \phi}^{\alpha, \beta, \gamma}(u_1, u_2, u_3, u_4). \quad (189)$$

where

$$X_{\delta, \epsilon, \phi}^{\alpha, \beta, \gamma}(u_1, u_2, u_3, u_4) := \chi_{\alpha}(u_1^{-1} u_2) \chi_{\beta}(u_4^{-1} u_2) \chi_{\gamma}(u_2^{-1} u_3) \chi_{\delta}(u_4^{-1} u_3) \chi_{\epsilon}(u_3^{-1} u_1) \chi_{\phi}(u_1^{-1} u_4) \quad (190)$$

Note that when $n = 2$, ??? recovers the well-known formula for the square of the 6j-symbol for $SU(n)$ (see [FL03, Equation (1)] and also [Bar98; FK00]).

Proof of ???. The proof follows from (i) the following identities concerning the character $\chi_{\lambda}(u) = \text{Tr}(\pi_{\lambda}(u))$ and duality $\pi_{\lambda}^*(u) = (\pi_{\lambda}(u^{-1}))^T$ relations

$$\chi_{\lambda}(u) = \boxed{\pi_{\lambda}(u)} \downarrow \lambda, \quad \pi_{\gamma}^*(u) = \boxed{\pi_{\gamma}^*(u)} \downarrow \gamma = \boxed{\pi_{\gamma}(u^{-1})} \downarrow \gamma, \quad (191)$$

and (ii) the integral formula for the projection operator $\Pi_{\alpha,\beta}^\gamma$ onto the subspace of invariant vectors in $V_\alpha \otimes V_\beta \otimes V_\gamma^*$:

$$\begin{array}{c} \alpha \quad \beta \quad \gamma \\ \uparrow \quad \uparrow \quad \downarrow \\ \boxed{\Pi_{\alpha,\beta}^\gamma} \\ \downarrow \quad \downarrow \quad \downarrow \\ \alpha \quad \beta \quad \gamma \end{array} = \begin{array}{c} \alpha \quad \beta \quad \gamma \\ \curvearrowright \quad \curvearrowright \quad \curvearrowright \\ C_{\alpha,\beta}^\gamma \\ \curvearrowleft \quad \curvearrowleft \quad \curvearrowleft \\ \alpha \quad \beta \quad \gamma \end{array} = \int du \begin{array}{c} \alpha \quad \beta \quad \gamma \\ \uparrow \quad \uparrow \quad \downarrow \\ \boxed{\pi_\alpha(u)} \quad \boxed{\pi_\beta(u)} \quad \boxed{\pi_\gamma^*(u)} \\ \downarrow \quad \downarrow \quad \downarrow \\ \alpha \quad \beta \quad \gamma \end{array}. \quad (192)$$

Applied four times, we obtain

$$\begin{aligned} \begin{array}{c} \textcircled{L_\epsilon}^2 C_{\alpha,\phi}^\epsilon C_{\alpha,\beta}^\gamma C_{\gamma,\delta}^\epsilon C_{\beta,\delta}^\phi = \end{array} & \begin{array}{c} \begin{array}{c} \left\{ \begin{array}{ccc} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{array} \right\}^\dagger \\ \uparrow \quad \uparrow \quad \uparrow \\ \begin{array}{c} \left\{ \begin{array}{ccc} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{array} \right\} \end{array} \\ \downarrow \quad \downarrow \quad \downarrow \end{array} \end{array} \\ & \begin{array}{c} \begin{array}{c} \phi \quad \epsilon \quad \alpha \quad \alpha \quad \beta \quad \gamma \quad \gamma \quad \delta \quad \epsilon \quad \delta \quad \beta \quad \phi \\ \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \\ C_{\alpha,\phi}^\epsilon \quad C_{\alpha,\beta}^\gamma \quad C_{\gamma,\delta}^\epsilon \quad C_{\beta,\delta}^\phi \end{array} \\ \begin{array}{c} \phi \quad \epsilon \quad \alpha \quad \alpha \quad \beta \quad \gamma \quad \gamma \quad \delta \quad \epsilon \quad \delta \quad \beta \quad \phi \\ \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \\ C_{\alpha,\phi}^\epsilon \quad C_{\alpha,\beta}^\gamma \quad C_{\gamma,\delta}^\epsilon \quad C_{\beta,\delta}^\phi \end{array} \end{array} \\ & = \Pi_i \int du_i \begin{array}{c} \begin{array}{c} \bar{u}_1 \quad u_1 \quad \bar{u}_1 \quad u_2 \quad u_2 \quad \bar{u}_2 \quad u_3 \quad u_3 \quad \bar{u}_3 \quad u_4 \quad u_4 \quad \bar{u}_4 \\ \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \\ \phi \quad \epsilon \quad \alpha \quad \alpha \quad \beta \quad \gamma \quad \gamma \quad \delta \quad \epsilon \quad \delta \quad \beta \quad \phi \end{array} \end{array} \\ & = \Pi_i \int du_i \begin{array}{c} \begin{array}{c} \bar{u}_1 \quad u_2 \quad \bar{u}_2 \quad u_3 \quad u_3 \quad \bar{u}_4 \quad u_2 \quad \bar{u}_4 \quad u_1 \quad \bar{u}_3 \quad \bar{u}_1 \quad u_4 \\ \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \\ \alpha \quad \gamma \quad \delta \quad \beta \quad \epsilon \quad \phi \end{array} \end{array}. \quad (193) \end{aligned}$$

□

Remark 58. With the integral character formula $R_{\delta\epsilon\phi}^{\alpha\beta\gamma}$ established, it is interesting to compare it against the character formula for the heat kernel of a compact Lie group G [Feg78], denoted by $K_G(g_1, g_2; t)$ in [McS21]:

$$K_G(g_1, g_2; t) = \sum_{\lambda} \dim V_{\lambda}^n \chi_{\lambda}(g_1 g_2^{-1}) e^{-t(|\lambda + \rho|^2 - |\rho|^2)/2}, \quad (194)$$

where the sum is over all dominant integral weights λ for G and ρ the associated Weyl vector. It is conceivable that there might be a method for expressing $R_{\delta\epsilon\phi}^{\alpha\beta\gamma}$ as a low temperature limit, $t \rightarrow 0$, of an integral over products of heat kernels.

7.2 Littlewood-Richardson coefficients & the trigonal Horn problem

TC: TODO: There is a lot of work left to do to clean up this section. In particular, it would be very useful to present a brief pedagogical account of the connection between the Littlewood-Richardson coefficients and the trigonal Horn problem and then use that intuition as a basis for extrapolating to the case of the tetrahedral Horn problem and the need for a representation-theory object which captures the additional consistency constraints imposed by the tetrahedral Horn problem. TODO: This section also needs a lot of work,

there are about half a dozen or so ways of building up the connection between the Littlewood-Richardson coefficients and the trigonal Horn problem. In the end, the approach we need to take here should mirror the approach in [Chr08] as closely as possible because it is this approach that we make use of when generalizing from the trigonal to the tetrahedral Horn problem. For example, one common strategy for simplifying the presentation is to shift from the asymmetric variant of the Horn problem where $A + B = C$ to the symmetric variant of the Horn problem where $A + B + C = 0$. While this makes things conceptually much clearer, when generalizing to the tetrahedral Horn problem, the more “symmetric” approach becomes a slight obstacle and forces one to consider artificial matching conditions of the form $C + C' = 0$ and therefore to consider a total of twelve matrices.

It is also worth mentioning that one can describe the Horn inequalities themselves in terms of the LR coefficients. This discussion should be included in Section 5.1.

Our first result can be viewed as a simple condition which is sufficient to conclude that a given triple of integer eigenvalues are trigonal.

Lemma 59. *If $c_{\alpha,\beta}^\gamma > 0$, then there exists Hermitian matrices $A \in \mathcal{O}_\alpha$, $B \in \mathcal{O}_\beta$, and $C \in \mathcal{O}_\gamma$, satisfying $A + B - C = 0$.*

Proof. Our proof proceeds in the style of [BVW18, Proposition 2.3]. Given any dominant weight $\lambda \in \mathbb{Z}_{\text{dec}}^n$, let P_λ denote the Hermitian projection operator onto the invariant $\square$

Using the theory of spectrum estimation developed in the context of quantum information theory [ARS88; KW01; HM02], the following pair of theorems were proven in [Chr08]. Recall that $(\alpha, \beta, \gamma) \in (\mathbb{R}_{\text{dec}}^n)^{\times 3}$ are said to be *trigonal* if there exists Hermitian matrices $A, B, C \in \mathcal{H}(n)$ satisfying (i) $A + B + C = 0$ and (ii)

$$(\text{eig}(A), \text{eig}(B), \text{eig}(C)) = (\alpha, \beta, \gamma). \quad (195)$$

Theorem 60. *If $c_{\alpha,\beta,\gamma} > 0$ for some $\alpha, \beta, \gamma \in \mathbb{Z}_{\text{dec}}^n$, then (α, β, γ) are trigonal.*

Theorem 61. *If $(\alpha, \beta, \gamma) \in (\mathbb{R}_{\text{dec}}^n)^{\times 3}$ are triangular, then there exists a graded sequence $(\tilde{\alpha}(n), \tilde{\beta}(n), \tilde{\gamma}(n)) \in (\mathbb{Z}_{\text{dec}}^n)^{\times 3}$ such that (i) for all $n \in \mathbb{N}_{>0}$,*

$$c_{\tilde{\alpha}(n), \tilde{\beta}(n), \tilde{\gamma}(n)} > 0, \quad (196)$$

and (ii)

$$\lim_{n \rightarrow \infty} \frac{\tilde{\alpha}(n)}{n} = \alpha, \quad \lim_{n \rightarrow \infty} \frac{\tilde{\beta}(n)}{n} = \beta, \quad \lim_{n \rightarrow \infty} \frac{\tilde{\gamma}(n)}{n} = \gamma. \quad (197)$$

Proposition 62 (Kempf-Ness). *Let $(A, B, C) \in \mathcal{O}_\alpha \times \mathcal{O}_\beta \times \mathcal{O}_\gamma$ be a triple of $d \times d$ Hermitian matrices with integer-valued eigenvalues and let*

$$w_{A,B,C} := \varphi(A) \otimes \varphi(B) \otimes \varphi(C) = v_\alpha^{u_A} \otimes v_\beta^{u_B} \otimes v_\gamma^{u_C} \quad (198)$$

be the associated coherent state of unit norm in $V_\alpha \otimes V_\beta \otimes V_\gamma$. Define the capacity of (A, B, C) as

$$\text{cap}(A, B, C) := \inf_{g \in \text{GL}(n)} \|[\pi_\alpha(g) \otimes \pi_\beta(g) \otimes \pi_\gamma(g)]w_{A,B,C}\|^2, \quad (199)$$

$$= \inf_{g \in \text{GL}(n)} \left\| v_\alpha^{gu_A} \otimes v_\beta^{gu_B} \otimes v_\gamma^{gu_C} \right\|^2. \quad (200)$$

Then $\text{cap}(A, B, C) \in [0, 1]$ with

$$\text{cap}(A, B, C) = 1 \iff \mu(A, B, C) = A + B + C = 0. \quad (201)$$

Proof. Summarize the direct and pedagogical Kempf-Ness style proof in [BVW18, Proposition 2.3]. $\square$

The following is a combination of [FW20, Theorem 1 & Proposition 3.1] (using the convention where capacities and constant portions are defined in terms of squared norms instead of norms directly).

Theorem 63 (Strong-duality & Occasionality). *Let everything be as in ?? and let k be a positive integer. Define the constant portion of A, B, C of degree k to be*

$$\text{cst}_k(A, B, C) = \left\| \Gamma_{\alpha, \beta, \gamma}^k w_{A, B, C}^{\otimes k} \right\|^{\frac{2}{k}}, \quad (202)$$

where $\Gamma_{\alpha, \beta, \gamma}^k$ is the projection operator onto the subspace of $\text{GL}(n)$ -invariant vectors in $(V_\alpha \otimes V_\beta \otimes V_\gamma)^{\otimes k}$. Then

$$\text{cap}(A, B, C) = \sup_k \text{cst}_k(A, B, C) \quad (203)$$

Moreover, when $\text{cap}(A, B, C) = 1$ and $\mu(A, B, C) = 0$, there exists an integer $m > 0$ such that

$$\lim_{k \rightarrow \infty} (mk)^s \text{cst}_{mk}(A, B, C) > 0 \quad (204)$$

where $s = \dim U(n) - \dim U(n)_{[w_{A, B, C}]}$.

Corollary 64. *Consider the quantity $\tilde{c}_{\alpha, \beta, \gamma}$ which measures the fraction of vectors $V_\alpha \otimes V_\beta \otimes V_\gamma$ which are $\text{GL}(n)$ -invariant:*

$$\tilde{c}_{\alpha, \beta, \gamma} = \frac{\dim[(V_\alpha \otimes V_\beta \otimes V_\gamma)^{\text{GL}(n)}]}{\dim(V_\alpha \otimes V_\beta \otimes V_\gamma)}. \quad (205)$$

If there exists $(A, B, C) \in \mathcal{O}_\alpha \times \mathcal{O}_\beta \times \mathcal{O}_\gamma$ such that $A + B + C = 0$, then there exists an integer $K > 0$ such that $c_{K\alpha, K\beta, K\gamma} \geq 1^8$, and therefore for all integers $M > 0$

$$\tilde{c}_{MK\alpha, MK\beta, MK\gamma} \geq \frac{1}{\dim(V_{MK\alpha} \otimes V_{MK\beta} \otimes V_{MK\gamma})}. \quad (206)$$

On the other hand, if there does not exist $(A, B, C) \in \mathcal{O}_\alpha \times \mathcal{O}_\beta \times \mathcal{O}_\gamma$ such that $A + B + C = 0$, then the function $k \mapsto \tilde{c}_{k\alpha, k\beta, k\gamma}$ admits an exponential upper bound⁹

$$\tilde{c}_{k\alpha, k\beta, k\gamma} \leq \max_{A, B, C} \text{cap}(A, B, C)^k. \quad (207)$$

Proof. The lower-bound follows from the operator inequality

$$|v_\lambda^u\rangle \langle v_\lambda^u| \leq \text{id}_{V_\lambda}, \quad (208)$$

for all unit vectors $v_\lambda^u \in V_\lambda$ while the upper-bound follows from the resolution of identity over the normalized Haar measure du for $U(n)$:

$$\text{id}_{V_\lambda} = \dim(V_\lambda) \int_{u \in U(n)} du |v_\lambda^u\rangle \langle v_\lambda^u|. \quad (209)$$

Applying ?? three times to the definition of $\tilde{c}_{k\alpha, k\beta, k\gamma}$ yields

$$\begin{aligned} \tilde{c}_{k\alpha, k\beta, k\gamma} &= \frac{\dim[(V_{k\alpha} \otimes V_{k\beta} \otimes V_{k\gamma})^{\text{GL}(n)}]}{\dim(V_{k\alpha} \otimes V_{k\beta} \otimes V_{k\gamma})} \\ &= \int du_A du_B du_C \langle (v_{k\alpha}^{u_A} \otimes v_{k\beta}^{u_B} \otimes v_{k\gamma}^{u_C}), \Gamma_{k\alpha, k\beta, k\gamma}^1 (v_{k\alpha}^{u_A} \otimes v_{k\beta}^{u_B} \otimes v_{k\gamma}^{u_C}) \rangle. \end{aligned} \quad (210)$$

While the operator $\Gamma_{k\alpha, k\beta, k\gamma}^1$ appearing in ?? above projects onto the subspace of $\text{GL}(n)$ -invariant vectors in $V_{k\alpha} \otimes V_{k\beta} \otimes V_{k\gamma}$, the operator $\Gamma_{\alpha, \beta, \gamma}^k$ projects onto the subspace of $\text{GL}(n)$ -invariant vectors in $(V_\alpha \otimes V_\beta \otimes V_\gamma)^{\otimes k}$. Since $V_{k\lambda}$ always embeds isometrically and equivariantly into $V_\lambda^{\otimes k}$ for any λ , we have that $V_{k\alpha} \otimes V_{k\beta} \otimes V_{k\gamma}$ is a subspace of $(V_\alpha \otimes V_\beta \otimes V_\gamma)^{\otimes k}$ with respect to a $\text{GL}(n)$ -equivariant isometry and therefore the projection operators satisfy

$$\Gamma_{k\alpha, k\beta, k\gamma}^1 = (\iota_\alpha^k \otimes \iota_\beta^k \otimes \iota_\gamma^k)^\dagger \Gamma_{\alpha, \beta, \gamma}^k (\iota_\alpha^k \otimes \iota_\beta^k \otimes \iota_\gamma^k). \quad (211)$$

⁸In fact the proof of the saturation conjecture implies $K = 1$ always suffices.

⁹The semigroup property of Littlewood-Richardson coefficients together with the at most polynomial growth in $\dim(V_{k\lambda})$ combined with the exponential upper bound on $\tilde{c}_{k\alpha, k\beta, k\gamma}$ actually implies $c_{\alpha, \beta, \gamma} = 0$ whenever $A + B + C = 0$.

Moreover, using the $\text{GL}(n)$ -equivariance of $\Gamma_{\alpha,\beta,\gamma}^k$ and the inequality $\Gamma_{\alpha,\beta,\gamma}^k \leq (\text{id}_{V_\alpha \otimes V_\beta \otimes V_\gamma})^{\otimes k}$ yields, for all $g \in \text{GL}(n)$, an upper bound on the integrand in ??:

$$\langle (v_{k\alpha}^{u_A} \otimes v_{k\beta}^{u_B} \otimes v_{k\gamma}^{u_C}), \Gamma_{k\alpha,k\beta,k\gamma}^1(v_{k\alpha}^{u_A} \otimes v_{k\beta}^{u_B} \otimes v_{k\gamma}^{u_C}) \rangle \quad (212)$$

$$= \langle (v_\alpha^{u_A} \otimes v_\beta^{u_B} \otimes v_\gamma^{u_C})^{\otimes k}, \Gamma_{\alpha,\beta,\gamma}^k(v_\alpha^{u_A} \otimes v_\beta^{u_B} \otimes v_\gamma^{u_C})^{\otimes k} \rangle, \quad (213)$$

$$= \langle (v_\alpha^{gu_A} \otimes v_\beta^{gu_B} \otimes v_\gamma^{gu_C})^{\otimes k}, \Gamma_{\alpha,\beta,\gamma}^k(v_\alpha^{gu_A} \otimes v_\beta^{gu_B} \otimes v_\gamma^{gu_C})^{\otimes k} \rangle, \quad (214)$$

$$\leq \langle (v_\alpha^{gu_A} \otimes v_\beta^{gu_B} \otimes v_\gamma^{gu_C})^{\otimes k}, (v_\alpha^{gu_A} \otimes v_\beta^{gu_B} \otimes v_\gamma^{gu_C})^{\otimes k} \rangle, \quad (215)$$

$$= \left\| v_\alpha^{gu_A} \otimes v_\beta^{gu_B} \otimes v_\gamma^{gu_C} \right\|^{2k}. \quad (216)$$

Finally, let (u'_A, u'_B, u'_C) be unitaries which achieve the maximum of the integrand appearing in ?? (continuity of the integrand and compactness of $\text{U}(n)$ guarantees existence) such that $\tilde{c}_{k\alpha,k\beta,k\gamma}$ admits the bound

$$\tilde{c}_{k\alpha,k\beta,k\gamma} \leq \left\| v_\alpha^{gu'_A} \otimes v_\beta^{gu'_B} \otimes v_\gamma^{gu'_C} \right\|^{2k}. \quad (217)$$

Since $g \in \text{GL}(n)$ is arbitrary and (u'_A, u'_B, u'_C) is independent of g by definition one obtains

$$\tilde{c}_{k\alpha,k\beta,k\gamma} \leq \inf_{g \in \text{GL}(n)} \left\| v_\alpha^{gu'_A} \otimes v_\beta^{gu'_B} \otimes v_\gamma^{gu'_C} \right\|^{2k} = \text{cap}(A', B', C')^k \leq \max_{A,B,C} \text{cap}(A, B, C)^k. \quad (218)$$

TODO: Prove that the capacity is attained by some Hermitian matrices A, B, C (maybe using the quantitative Kempf-Ness theorem [Bür+19, Theorem 1.17]). This establishes the claimed upper bound. To prove the claimed lower bound, let $v_\alpha^{u'_A} \otimes v_\beta^{u'_B} \otimes v_\gamma^{u'_C}$ be the coherent state associated with the Hermitian matrices (A, B, C) satisfying $A + B + C = 0$. Then using ?? and the positive demidefiniteness of $\Gamma_{k\alpha,k\beta,k\gamma}^1$ we obtain,

$$\begin{aligned} \tilde{c}_{k\alpha,k\beta,k\gamma} &= \frac{\dim[(V_{k\alpha} \otimes V_{k\beta} \otimes V_{k\gamma})^{\text{GL}(n)}]}{\dim(V_{k\alpha} \otimes V_{k\beta} \otimes V_{k\gamma})} \\ &\geq \frac{\langle (v_{k\alpha}^{u'_A} \otimes v_{k\beta}^{u'_B} \otimes v_{k\gamma}^{u'_C}), \Gamma_{k\alpha,k\beta,k\gamma}^1(v_{k\alpha}^{u'_A} \otimes v_{k\beta}^{u'_B} \otimes v_{k\gamma}^{u'_C}) \rangle}{\dim(V_{k\alpha} \otimes V_{k\beta} \otimes V_{k\gamma})}, \\ &= \frac{\langle (v_\alpha^{u'_A} \otimes v_\beta^{u'_B} \otimes v_\gamma^{u'_C})^{\otimes k}, \Gamma_{\alpha,\beta,\gamma}^k(v_\alpha^{u'_A} \otimes v_\beta^{u'_B} \otimes v_\gamma^{u'_C})^{\otimes k} \rangle}{\dim(V_{k\alpha} \otimes V_{k\beta} \otimes V_{k\gamma})}, \\ &= \frac{\text{cst}_k(A, B, C)}{\dim(V_{k\alpha} \otimes V_{k\beta} \otimes V_{k\gamma})}. \end{aligned} \quad (219)$$

Now ?? implies there exists positive integers $m, s > 0$ such that for sufficiently large k , $(mk)^s \text{cst}_{mk}(A, B, C) > 0$ which implies $c_{mk\alpha,mk\beta,mk\gamma} \neq 0$ and thus $c_{mk\alpha,mk\beta,mk\gamma} \geq 1$. Taking $K = mk$ for sufficiently large k proves the claim. $\square$

A Multinomial distribution inequalities

In this section, we recall a handful of identities and inequalities concerning multinomial coefficients and the multinomial distribution in the context of Sanov's theorem [DZ10, Chapter 2]. Let $q = (q_1, \dots, q_n)$ be a probability distribution such that for all $i \in \{1, \dots, n\}$, kq_i is an integer. Let $M_k(q)$ denote the following multinomial coefficient:

$$M_k(q) := \binom{k}{kq_1, \dots, kq_n} = \frac{k!}{(kq_1)! \dots (kq_n)!}, \quad \text{\texttt{eq:multinomial_coefficient}} \quad (220)$$

which can be bounded (see [DZ10, Lemma 2.1.8]) in terms of $H(q)$ by ,

$$(k+1)^{-n} e^{kH(q)} \leq M_k(q) \leq e^{kH(q)}, \quad \text{\texttt{eq:multinomial_coefficient_bound}} \quad (221)$$

where $H(q) = -\sum_{i=1}^n q_i \log q_i$ is the Shannon entropy of q . Additionally, we have for any probability distribution $p = (p_1, \dots, p_n)$,

$$p_1^{kq_1} \dots p_n^{kq_n} = e^{-k(H(q) + D(q\|p))} \quad \text{\texttt{eq:multinomial_probability}} \quad (222)$$

where $D(q\|p) = \sum_{i=1}^n q_i \log q_i/p_i$ denotes the Kullback-Leibler divergence or relative entropy of q from p with the convention that $D(q\|p) = \infty$ if $q_i > 0$ whenever $p_i = 0$. Note that $D(q\|p) = 0$ if and only if $p = q$. Combining Eq. (221) with Eq. (222) places bounds on the multinomial distribution

$$(k+1)^{-n} e^{-kD(q\|p)} \leq \binom{k}{kq_1, \dots, kq_n} p_1^{kq_1} \dots p_n^{kq_n} \leq e^{-kD(q\|p)} \quad (223)$$

If needed, it is possible to relax these bounds further using Pinsker's inequality for $D(q\|p)$ in conjunction with its reverse (which only holds for $D(q\|p) < \infty$) [GSS19]:

$$2d_{\text{TV}}^2(p, q) \leq D(q\|p) \leq \frac{4}{p_{\min}} d_{\text{TV}}^2(p, q) \quad (224)$$

where

$$d_{\text{TV}}(p, q) = \max_{1 \leq j \leq n} |p_j - q_j| = \frac{1}{2} \sum_{j=1}^n |p_j - q_j| \quad \text{\texttt{eq:pinskera}} \quad (225)$$

is known as the *total variation distance* between p and q . Now for any distribution $p = (p_1, \dots, p_n)$ and any positive integer k , it is possible to find another distribution $p' = (p'_1, \dots, p'_n)$ such that (i) kp'_i is an integer for all $i \in \{1, \dots, n\}$, (ii) $D(p'\|p) < \infty$ is finite, and (iii) p' approximates p in the sense that for some constant $C \geq 0$,

$$d_{\text{TV}}(p, p') \leq \frac{C}{k}. \quad \text{\texttt{eq:tv_approx}} \quad (226)$$

Concretely, one can choose p' such that

$$kp'_j := \text{nint} \left(k \sum_{i=1}^j p_i \right) - \text{nint} \left(k \sum_{i=1}^{j-1} p_i \right) \quad \text{\texttt{eq:nint_approx}} \quad (227)$$

where $\text{nint} : \mathbb{R} \rightarrow \mathbb{Z}$ is the function which rounds each real number to the nearest integer (with half integers always rounded up). Using this approximation, we have $|kp_j - kp'_j| \leq 1$ because each rounding operation in Eq. (227) introduces an error of at most $1/2$ and thus Eq. (226) holds with $C = 1$. In any case, when Eq. (226) holds for some constant $C \geq 0$, Eq. (225) reveals $D(p'\|p) \leq \frac{4C}{p_{\min} k^2}$ which yields a useful, albeit crude, lower bound for the multinomial distribution:

$$\binom{k}{kp'_1, \dots, kp'_n} p_1^{kp'_1} \dots p_n^{kp'_n} \geq \exp \left(-\frac{4C^2}{p_{\min} k} - n \log(k+1) \right). \quad \text{\texttt{lem:multinomial_dist_lower_bound}} \quad (228)$$

Lemma 65. Let $n \geq 2$ and let $(w_1, \dots, w_n) \in \mathbb{R}_{>0}^n$ have minimum $w_{\min} := \min\{w_1, \dots, w_n\} > 0$ and let k be a positive integer satisfying $k \geq w_{\min}^{-1}$ and define positive integers $(z_1, \dots, z_n) \in \mathbb{N}_{>0}^n$ by the equation

$$z_i := \lceil kw_i \rceil. \quad (229)$$

Let $Z = z_1 + \dots + z_n$ and $W = w_1 + \dots + w_n$ and defined probability distributions $\underline{z}$ and $\underline{w}$ by

$$\underline{z}_i = \frac{z_i}{Z}, \quad \text{and} \quad \underline{w}_i = \frac{w_i}{W}, \quad (230)$$

for all $i \in \{1, \dots, n\}$. Then

$$\frac{Z!}{z_1! \dots z_n!} \frac{w_1^{z_1} \dots w_n^{z_n}}{W^Z} \geq e^{-4(n-1)^2} (kW + n + 1)^{-n} \quad \text{\textcolor{blue}{\code{eq:multinomial_dist_lower_bound}}} \quad (231)$$

Proof. Let $\delta_j := z_j - kw_j$ denote the rounding surplus such that $0 \leq \delta_j < 1$. Note that for $k \gg n$, we have $Z \approx kW$ because

$$kW \leq Z < kW + n. \quad (232)$$

In this regime, we wish to show that $\underline{z}_i \approx \underline{w}_i$.

$$|\underline{z}_i - \underline{w}_i| = \left| \frac{z_i}{Z} - \frac{w_i}{W} \right| = \frac{|Wz_i - Zw_i|}{ZW}. \quad (233)$$

The numerator above can be bounded independently of k by,

$$|Wz_i - Zw_i| \quad (234)$$

$$= |(kw_i + \delta_i)(w_1 + \dots + w_n) - w_i(\delta_1 + kw_1 + \dots + \delta_n + kw_n)|, \quad (235)$$

$$= \left| \delta_i(w_1 + \dots + \hat{w}_i + \dots + w_n) - w_i(\delta_1 + \dots + \hat{\delta}_i + \dots + \delta_n) \right|, \quad (236)$$

$$\leq |\delta_i(w_1 + \dots + \hat{w}_i + \dots + w_n)| + \left| w_i(\delta_1 + \dots + \hat{\delta}_i + \dots + \delta_n) \right|, \quad (237)$$

$$\leq (w_1 + \dots + \hat{w}_i + \dots + w_n) + (n-1)(w_i), \quad (238)$$

$$\leq (n-1)(w_1 + \dots + w_n), \quad (239)$$

$$= (n-1)W. \quad (240)$$

Consequently,

$$d_{\text{TV}}(\underline{z}, \underline{w}) = \max_{1 \leq i \leq n} |\underline{z}_i - \underline{w}_i| \leq \frac{n-1}{Z} \quad (241)$$

and therefore

$$\frac{Z!}{z_1! \dots z_n!} \frac{w_1^{z_1} \dots w_n^{z_n}}{W^Z} \geq (Z+1)^{-n} \exp(-ZD(\underline{z} \parallel \underline{w})), \quad (242)$$

$$\geq (Z+1)^{-n} \exp \left[-\frac{4ZW}{w_{\min}} \left(\frac{n-1}{Z} \right)^2 \right], \quad (243)$$

$$\geq (kW + n + 1)^{-n} \exp \left[-\frac{4(n-1)^2}{kw_{\min}} \right], \quad (244)$$

$$\geq (kW + n + 1)^{-n} \exp(-4(n-1)^2). \quad (245)$$

□

The following is an alternative lower bound on the multinomial distribution based on the Stirling approximation which would work just as well in the main text. TODO: Figure out the best lower bound on the multinomial distribution to use here.

\code{lem:multinomial_dist_lower_bound_alt}

Lemma 66. *Let everything be as in Lemma 65. Then*

$$\frac{Z!}{z_1! \dots z_n!} \frac{w_1^{z_1} \dots w_n^{z_n}}{W^Z} \geq \left(\frac{1}{20} \right)^n \left(\frac{W}{w_1 \dots w_n} \right)^{\frac{1}{2}} \frac{1}{k^{\frac{3n-1}{2}}} \quad \text{\textcolor{blue}{\code{eq:multinomial_dist_lower_bound_alt}}} \quad (246)$$

Proof. The proof follows from a version of Stirling's approximation due to Robbins [Rob55] which states

$$k! = \sqrt{2\pi k} k^{k+\frac{1}{2}} e^{-k} e^{r_k}, \quad \text{where} \quad \frac{1}{12k+1} \leq r_k \leq \frac{1}{12k}. \quad (247)$$

Therefore,

$$\frac{Z!}{z_1! \cdots z_n!} \frac{w_1^{z_1} \cdots w_n^{z_n}}{W^Z} = \frac{Z!}{W^Z} \frac{w_1^{z_1}}{z_1!} \cdots \frac{w_n^{z_n}}{z_n!} \quad (248)$$

$$= \left(\frac{1}{2\pi}\right)^{\frac{n-1}{2}} \left(\frac{W}{w_1 \cdots w_n}\right)^{\frac{1}{2}} \left(\frac{Z}{W}\right)^{Z+\frac{1}{2}} \left(\frac{w_1}{z_1}\right)^{z_1+\frac{1}{2}} \cdots \left(\frac{w_n}{z_n}\right)^{z_n+\frac{1}{2}} \exp\left(r_Z - \sum_i r_{z_i}\right). \quad (249)$$

In this setting, we have assumed $kw_i \geq 1$ which implies

$$kw_i \leq z_i < kw_i + 1 \leq 2kw_i, \quad \text{and} \quad kW \leq Z < kW + n < 2kW. \quad (250)$$

Consequently¹⁰,

$$\left(\frac{Z}{W}\right)^{Z+\frac{1}{2}} \geq k^{kW+\frac{1}{2}}, \quad \left(\frac{w_i}{z_i}\right)^{z_i+\frac{1}{2}} \geq \left(\frac{w_i}{kw_i+1}\right)^{kw_i+\frac{3}{2}} \geq \frac{2^{-\frac{5}{2}}}{k^{kw_i+\frac{3}{2}}}. \quad (251)$$

and the remainder term satisfies

$$r_Z - \sum_i r_{z_i} \geq \frac{1}{12Z+1} - \sum_i \frac{1}{12z_i} \geq -\frac{n}{12}. \quad (252)$$

Altogether, we have

$$\frac{Z!}{z_1! \cdots z_n!} \frac{w_1^{z_1} \cdots w_n^{z_n}}{W^Z} = \left(\frac{1}{2\pi}\right)^{\frac{n-1}{2}} \left(\frac{W}{w_1 \cdots w_n}\right)^{\frac{1}{2}} \frac{2^{-\frac{5n}{2}}}{k^{\frac{3n-1}{2}}} \exp\left(-\frac{n}{12}\right). \quad (253)$$

Moreover,

$$\left(\frac{1}{2\pi}\right)^{\frac{n-1}{2}} 2^{-\frac{5n}{2}} \exp\left(-\frac{n}{12}\right) \geq \left(\frac{1}{20}\right)^n \quad (254)$$

□

¹⁰Here we use the identity that $x \geq 1$ implies $\left(\frac{x}{x+1}\right)^{x+\frac{3}{2}} \geq 2^{-5/2}$

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